When Is Label-Free Adaptation Knowable?

Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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Abstract

Test-time adaptation can improve models on unlabeled target data, but the same objective that helps under one distribution shift can silently hurt under another. Existing work mainly strengthens adaptation mechanisms; we study a prior decision problem: can a system decide, without target labels, whether it should adapt at all?

We formalize adaptation benefit as the excess target risk of a frozen baseline over an adapted candidate and define three label-free regimes: *knowably helpful*, *knowably harmful*, and *unknowable*. We prove that when two target worlds induce identical observable evidence but opposite adaptation benefit, no label-free rule can be correct in both; abstention is then information-theoretically necessary. Under a risk-alignment assumption, we give a finite-sample adapt/freeze/abstain certificate controlling false adaptation and false freezing. Our main theoretical result is an exact benefit-sign frontier: adaptation benefit is identifiable from observables iff an observable margin exceeds the calibration-drift budget on the disagreement region.

We instantiate the framework as KGA, a mechanism-agnostic wrapper around Tent, EATA, and SAR. Across anomaly routing, CIFAR-10, CIFAR-10-C stress tests, and natural-shift audits, KGA follows theory: adapt in helpful regimes, freeze in harmful regimes, and abstain when evidence is insufficient. On a CIFAR-10-C stress grid, KGA beats always-adapt and always-freeze for Tent and EATA with zero false-adapt decisions, while tying SAR. At ImageNet scale, it also beats both policies on a harmful SAR stream. On Camelyon17, ImageNet-R, and Office-Home, KGA is conservative and avoids overclaiming wins. K-Bound is not universal; value is highest when harmful adaptation is frequent, detectable, and costly.

Scope. We report both positive and negative results and do not claim universal improvement; all numbers come from code in the public repository (§11). A detailed proof/experiment status note appears at the end of §1.

1 Introduction

Deployed models rarely encounter the same distribution they were trained on. Test-time adaptation (TTA) addresses this mismatch by updating a model on unlabeled target data, often through entropy minimization, sample filtering, or stability-aware updates. In favorable shifts, such updates can recover lost accuracy. In unfavorable shifts, however, the same unlabeled objective can silently degrade performance: the model may sharpen wrong predictions, adapt to label imbalance, or collapse under non-stationary streams. Because target labels are unavailable at deployment, the system often cannot tell whether adaptation is helping or hurting.

Most work on TTA asks how to design a better adaptation mechanism. This paper asks a prior question: should the system adapt at all? We study label-free adaptation as a decision problem. Given a frozen baseline, an adapted candidate, and an unlabeled target batch, the system must choose one of three actions: adapt when adaptation is knowably helpful, freeze when adaptation is knowably harmful, and abstain when the available evidence cannot identify the sign of the benefit.

This decision is constrained by identifiability. We show that there exist target worlds with identical label-free evidence but opposite adaptation benefit. In such cases, no label-free rule can certify the correct decision in both worlds; abstention is not merely conservative, but information-theoretically necessary. Conversely, when the label-free evidence is risk-aligned and the benefit estimate admits a valid uncertainty radius, a finite-sample certificate can safely route between adapt, freeze, and abstain while controlling false-adapt and false-freeze errors.

Our main theoretical result is an exact benefit-sign frontier. The sign of the adaptation benefit is identifiable from label-free observables precisely when an observable margin on the disagreement region exceeds a calibration-drift budget. This characterizes why common label-free heuristics can work in benign regimes and fail under drift: they implicitly assume that this drift is zero or negligible. We further show that no label-free-checkable assumption can remove this obstruction in full generality; the remaining ambiguity is an irreducible one-bit orientation choice, which can only be supplied through structural assumptions that are falsifiable but not verifiable.

We instantiate the framework as KGA, a mechanism-agnostic wrapper around existing adaptation candidates such as Tent, EATA, and SAR. KGA does not introduce a new adaptation objective. Instead, it estimates the label-free benefit of a candidate, attaches a calibrated uncertainty radius, and returns adapt, freeze, or abstain. This makes KGA a safety layer for TTA rather than a replacement for TTA methods.

Empirically, KGA behaves according to the theory. On a pre-registered CIFAR-10-C stress grid, where harmful adaptation is frequent and detectable, KGA beats both always-adapt and always-freeze for Tent and EATA while making zero false-adapt decisions, and it ties the collapse-resistant SAR. In online non-stationary CIFAR-10 streams, always-adapt collapses in the harm-dominated regime and always-freeze forgoes gains in the helpful regime, while KGA remains near-best across both. At ImageNet scale, KGA also improves over both trivial policies on a harmful SAR stream. On natural-shift audits such as Camelyon17, ImageNet-R, and Office-Home, the certificate is conservative: it reports safety behavior and abstains when the evidence structure is not sufficient for a valid policy win.

Contributions.

- We formalize label-free TTA as an adapt/freeze/abstain decision problem governed by the sign of adaptation benefit.
- We prove an impossibility result showing that some adaptation decisions are fundamentally unknowable from label-free evidence.
- We derive a finite-sample certificate that controls false adaptation and false freezing under risk-aligned evidence.
- We characterize an exact benefit-sign frontier based on observable disagreement margin and calibration drift.
- We validate the framework across anomaly routing, controlled mixed regimes, online TTA, CIFAR-10-C stress grids, ImageNet-scale corruptions, and natural-shift audits.

K-Bound does not claim universal improvement. When adaptation is already safe, a certificate can only tie or slightly trail always-adapt. When freezing is already optimal, it should match freeze rather than force an unnecessary update. Its distinctive value appears when harmful adaptation is frequent, detectable, and costly—the regime where blind adaptation is most dangerous.

2 Related work and positioning

We position K-Bound as a precise delta over five adjacent lines of work. In each case the distinction is the one the theory formalizes: between *improving* an unlabeled objective and *certifying the sign* of its benefit before acting.

2.1 Test-time adaptation methods

Entropy-based TTA updates a frozen model on unlabeled target data. **Tent** minimizes prediction entropy over the batch-norm affine parameters; **SHOT** combines information maximization with pseudo-labeling for source-free adaptation; **TTT** trains a self-supervised auxiliary task and updates it at test time; **EATA** adds entropy filtering of unreliable samples plus a Fisher penalty against forgetting; and **SAR** adds sharpness-aware updates with a reset that aborts incipient collapse. These methods make adaptation *more robust*, and several detect or suppress instability, but all commit to adapting: none emits a *pre-adaptation* certificate separating helpful, harmful, and unknowable evidence, and none can *abstain* on the adapt/freeze decision itself. K-Bound is mechanism-agnostic—it *wraps* any such candidate (we evaluate Tent/EATA/SAR). Tempora (Sreeram et al., 2026; arXiv:2602.06136) shows that unconstrained TTA rankings are not stable under temporal pressure: no fixed method dominates across latency/budget scenarios. K-Bound is complementary to that result by certifying, per condition, whether to adapt, freeze, or abstain.

2.2 Protected and guarded adaptation

A second line guards adaptation *during* the update—online entropy matching, or betting-style monitors that halt or dampen a single failure mode. This protects one direction (typically against collapse) but does not pose the three-way decision with an explicit *unknowable* region, nor certify *before* acting that adapting will help.

2.3 Label-free accuracy and error estimation

A third line predicts target *performance* without labels: **ATC** thresholds softmax confidence calibrated on source; **Agreement-on-the-Line** exploits the linear correlation between in- and out-of-distribution model *agreement* to extrapolate accuracy; **AETTA** estimates TTA accuracy from dropout-prediction disagreement. Estimating $R_T(f)$—or detecting that TTA has failed *post hoc*—is strictly different from certifying $\operatorname{sign}\big(R_T(f_0) - R_T(f_a)\big)$ *before* adapting with a controlled false-adapt rate, the object K-Bound bounds.

2.4 Unsupervised risk estimation and identifiability

The hardness of our problem is inherited from unsupervised risk estimation. **Steinhardt & Liang** estimate risk without labels only under a conditional-independence structure; **Ben-David et al.** bound target risk by source risk plus an $\mathcal{H}$-divergence; **Lipton et al.** (BBSE) correct label shift under an invertible confusion matrix. Each shows risk is identifiable only under structural assumptions and impossible in general—precisely the boundary our Theorem 2 makes quantitative. We weaken the target from a *risk* to the *sign of a difference* of risks, which is identifiable on a strictly larger set (the observable disagreement region, §5.5).

2.5 Selective prediction, conformal, and anytime-valid inference

Selective prediction (Geifman & El-Yaniv) abstains on individual *predictions*; we instead abstain on the *adaptation decision*. Our finite-sample certificate reuses split-conformal prediction (**Vovk et al.**; **Angelopoulos & Bates**) to build the radius ε, and its streaming form (Appendix E) is a testing-by-betting e-process in the safe-anytime-valid-inference tradition (**Shafer**; **Ramdas et al.**). **The surviving gap.** No prior method provides a single adapt/freeze/abstain certificate with an *explicit, proven* unknowable region and a false-adapt guarantee. Table 1 makes the gap precise.

Work	label-free?	adapts?	predicts benefit *before* adapting?	abstains on the decision?	false-adapt guarantee?
Tent	yes	yes	no	no	no
SHOT	yes	yes	no	no	no
EATA	yes	yes	partial (filtering)	no	no
SAR	yes	yes	stability only	no	no
AETTA	yes	monitors	post-hoc estimate	—	no
KGA (ours)	yes	wrapper	**yes**	**yes**	**yes**

Table 1: Positioning. Prior TTA adapts and (some) detect failure; none gives a *pre-adaptation* adapt/freeze/abstain certificate with a false-adapt bound.

2.6 Benchmarks, code, and reproducibility

TTA progress is benchmarked on standardized corruption suites—CIFAR-10/100-C and ImageNet-C (Hendrycks & Dietterich) and the RobustBench common-corruptions track—and the field increasingly emphasizes standardized, executable evaluation (RobustBench; DomainBed). Most TTA methods ship official code, but evaluations vary in batch regime, stream composition, and update budget—*precisely* the axis along which adaptation flips between helpful and harmful. We therefore release K-Bound as a *pre-registered, executable* artifact: a public repository with pinned environments, a one-command reproduction harness, auto-downloaded standard corruption data, fixed seeds, machine-checked theorem validators, and the full stratified result manifests behind every table and figure (§11). Our certificate builds on split-conformal prediction (Vovk et al.; Angelopoulos & Bates) and, for the streaming variant, testing-by-betting e-processes (Shafer; Ramdas et al.).

3 Problem setup

Source $P_S(X, Y)$ has labels at train/validation; target $P_T(X, Y)$ exposes only $P_T(X)$ at test time. Fix a loss ℓ, a frozen model f_0, and an adapted/gated candidate f_a. Target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)]$. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a), \qquad \Delta > 0 \iff \text{adapting helps.} \tag{1}$$

Observable evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable *without* target labels (entropy, confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, density-ratio diagnostics). The conditional benefit is $\Delta(z) = \mathbb{E}[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Z = z]$. The central question is when $\operatorname{sign}\Delta(z)$ is recoverable from Z alone.

Notation. f_0 is the frozen model and f_a the adapted/gated candidate; $\Delta = R_T(f_0) - R_T(f_a)$ is the population adaptation benefit and $\Delta(z)$ its value conditioned on evidence $Z{=}z$; $\widehat{\Delta}$ is a label-free estimator of Δ with confidence radius ε at miscoverage level α; $g(Z) \in \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$ is the decision rule; and $D = \{x : f_0(x) \neq f_a(x)\}$ is the observable disagreement region. *Regret-to-oracle* denotes $R(g) - R(g^\star)$ against the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$.

4 Definitions

Definition 1 (Observable risk alignment / Identifiability)**.** Z is *risk-aligned* if the sign of adaptation benefit $\operatorname{sign}\Delta$ is identifiable from its law (identifiability)—specifically, no two admissible worlds with opposite benefit signs induce the same evidence distribution. We treat conformal coverage of the benefit estimator separately as a property of the calibration sample (see Theorem 4).

Definition 2 (Regimes)**.** At evidence z: *knowably helpful* if Z certifies $\Delta(z) > 0$ (**adapt**); *knowably harmful* if Z certifies $\Delta(z) < 0$ (**freeze**); *unknowable* if the same Z is compatible with both signs (**abstain**).

What "label-free" means here. *Target-label-free* (our setting): no labels from the target/test stream are ever used. *Validation labels allowed*: labeled source and a held-out labeled validation/probe set may calibrate the benefit estimator $\widehat{\Delta}$ and the radius ε. *Target-unsupervised evidence*: the deployment batch contributes only X, predictions, entropy, confidence, drift and other label-free statistics Z. The online-TTA results that use a clean labelled validation probe (§7) are target-label-free in this sense—they use no *target* labels—not unsupervised of all labels.

5 Theory

We present the theory as four pillars rather than a long theorem catalog. **Pillar I (impossibility floor)** combines non-identifiability and its quantitative Le Cam form (Thms 1–2), plus the gate regret identity (Thm 3), to formalize when abstention is forced. **Pillar II (certificate + audit)** gives the finite-sample adapt/freeze/abstain rule with forced-abstention and sample-complexity corollaries (Thm 4, Cors 1–2); the legacy budget proposition stack (Props 6/9/11/13 in prior drafts) is now treated as corollaries of one common audit framework. **Pillar III (identifiable regimes)** gives the positive covariate-shift regime and disagreement-region sign characterization (Thm 5, Thms 6–22). **Pillar IV (rates and sequentiality)** is deferred to appendix-level companions, including the anytime certificate (Appendix E).

5.1 Result 1: The unknowable regime and a tight lower bound

Example. Two hospitals' scans induce identical confidence/entropy statistics, yet in one, adapting helps and in the other (a flipped label frequency) it badly hurts; no label-free rule can separate them, so the safe action is to abstain.

Theorem 1 (Non-identifiability; with witness)**.** *Let $g(Z) \in \{\textit{ADAPT}, \textit{FREEZE}, \textit{ABSTAIN}\}$ be any (possibly randomized) rule depending on label-free evidence Z. There exist two target worlds P_T^1, P_T^2 with* (i) $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$ *and* (ii) $\operatorname{sign}\Delta^1 = -\operatorname{sign}\Delta^2 \neq 0$. *For any such pair,* $\mathrm{Law}(g)$ *is identical across the two worlds; hence no label-free rule commits to the benefit-maximizing action in both, and the action minimizing the worst-case committal regret over $\{P_T^1, P_T^2\}$ is* ABSTAIN.

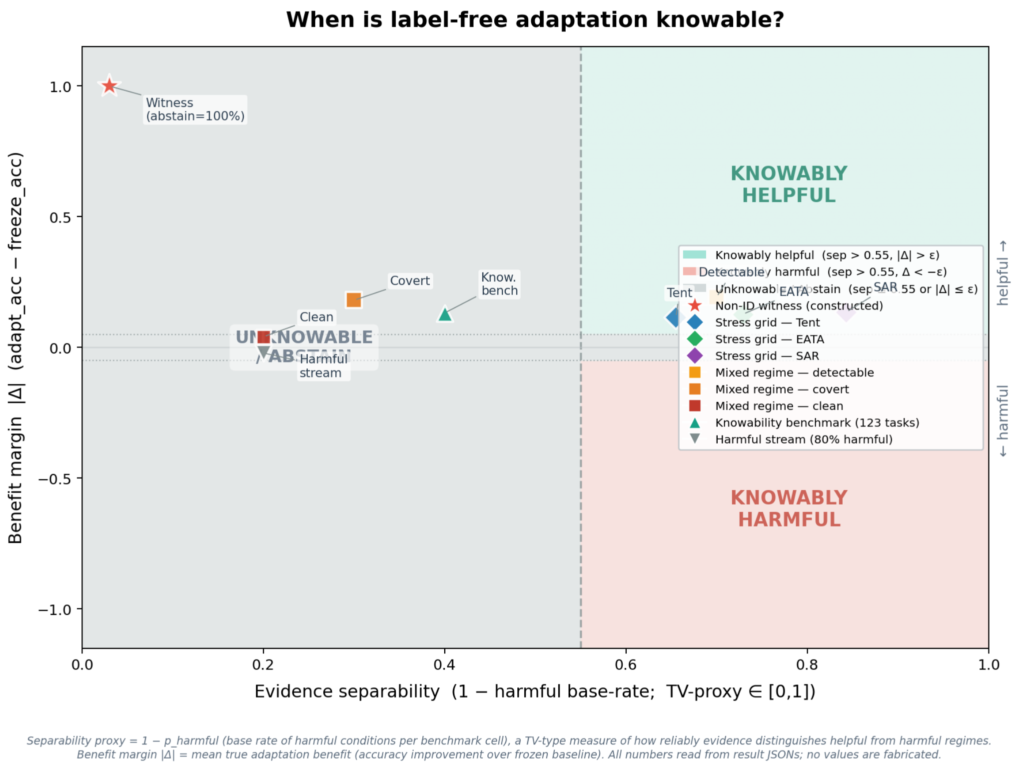

Figure 1: **The knowability phase diagram.** Each test-time condition lands at a point set by the evidence separability between opposite-benefit worlds (x-axis; a total-variation/detectability proxy) and the benefit margin $|\Delta|$ (y-axis). KGA commits (ADAPT/FREEZE) only outside the central *unknowable* wedge, where the certificate radius exceeds the estimated benefit ($\varepsilon > |\widehat{\Delta}|$); inside it the worst-case-optimal action is ABSTAIN (Theorem 1). Benchmark points—the CIFAR-10-C stress grid (Tent/EATA/SAR), the controlled mixed regime, and the clean non-identifiability witness—are overlaid from their measured values.

(Proof deferred to Appendix A.) Remark (weight). Theorem 1 is close to known non-identifiability of target risk without labels; its purpose here is to *justify the abstain action*, not to claim impossibility as novel. The quantitative strengthening below makes it a tight minimax statement.
Example. The more alike two such worlds look in the observable evidence (smaller TV distance), the closer any rule forced to commit is pushed toward a coin flip; the error floor vanishes only as their evidence separates.

Theorem 2 (Quantitative non-identifiability; Le Cam two-point lower bound)**.** *Consider two target worlds P_T^1, P_T^2 with $\Delta^1 < 0 < \Delta^2$. Let the label-free evidence be $Z = \phi(X_{1:n}, f_0, f_a)$ with n-sample evidence laws $P_{T,Z}^{j,(n)} := \mathrm{Law}(Z \mid P_T^j)$. For any (randomized) committal rule $g(Z) \in \{\text{ADAPT}, \text{FREEZE}\}$ define the mixed committal error $M(g) := \mathbb{P}_{P_T^1}[g = \text{ADAPT}] + \mathbb{P}_{P_T^2}[g = \text{FREEZE}]$. Then*

$$\inf_g M(g) \;=\; 1 - \mathrm{TV}(P_{T,Z}^{1,(n)}, P_{T,Z}^{2,(n)}) \;\geq\; 1 - \mathrm{TV}(P_T^{1,(n)}, P_T^{2,(n)}).$$

In the witness above (Z a function of X, $P_T^1(X) = P_T^2(X)$) the evidence TV is 0, so $\inf_g M(g) = 1$ for every n*: no committal rule beats blind guessing in the worst world, and under any positive coverage cost ABSTAIN is uniquely worst-case optimal. If the worlds become δ-detectable so the evidence laws*

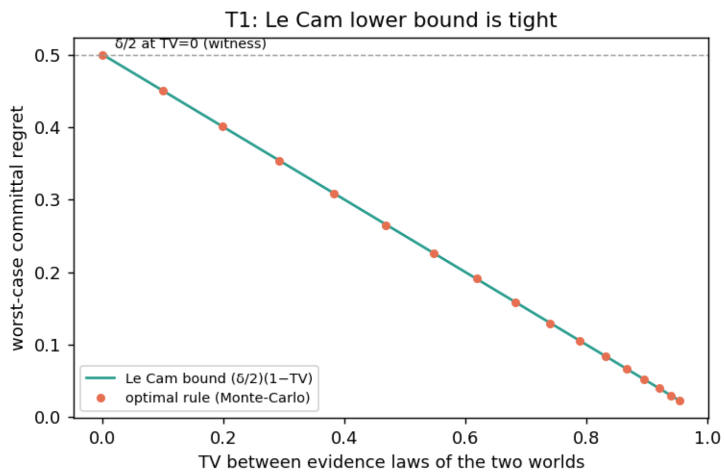

Figure 2: Theorem 2: the optimal committal rule's worst-case regret coincides with the Le Cam floor $\frac{\delta}{2}(1 - \mathrm{TV})$ across the TV sweep; at $\mathrm{TV} = 0$ (the witness) it equals $\delta/2$. Produced by `scripts/theory_extensions_validation.py`.

separate, $\inf_g M(g) = 1 - \mathrm{TV} \to 0$ *at the separation rate (for a location shift summarized by a sufficient statistic,* $\mathrm{TV} = 2\Phi(\frac{1}{2}\sqrt{n}\,\delta) - 1$*).*

(Proof deferred to Appendix A.) Numerical sanity check (`experiments/kbound/theory_validation/val_thm1_` in the witness the empirical $\inf_g M$ is ≈ 1 (KGA abstains 100%); in the detectable regime the empirical minimax error tracks the closed-form $1 - \mathrm{TV}$ across all n and shift magnitudes.

5.2 Result 2: The Bayes-optimal gate and plug-in regret

With $a_{\mathrm{FRZ}}(z) = \mathbb{E}[\ell(f_0(X), Y) \mid Z{=}z]$ and $a_{\mathrm{ADP}}(z) = \mathbb{E}[\ell(f_a(X), Y) \mid Z{=}z]$, so $\Delta(z) = a_{\mathrm{FRZ}}(z) - a_{\mathrm{ADP}}(z)$, a gate $g\colon z \mapsto \{0, 1\}$ has risk $R(g) = \mathbb{E}[g(Z)a_{\mathrm{ADP}}(Z) + (1 - g(Z))a_{\mathrm{FRZ}}(Z)]$. The Bayes gate is $g^\star(z) = \mathbf{1}[\Delta(z) > 0]$; a plug-in gate is $\widehat{g}(z) = \mathbf{1}[\widehat{\Delta}(z) > 0]$. *Example.* A gate's excess risk over the oracle equals the benefit it forfeits exactly where it picks the wrong sign—near-zero-benefit mistakes are cheap, large-benefit ones are not, which is why abstaining in the low-margin band is safe.

Theorem 3 (Plug-in regret decomposition)**.** *For every estimator* $\widehat{\Delta}$*,*

$$R(\widehat{g}) - R(g^\star) \;=\; \mathbb{E}\big[\,|\Delta(Z)|\,\mathbf{1}\{\operatorname{sign}\widehat{\Delta}(Z) \neq \operatorname{sign}\Delta(Z)\}\,\big],$$

where $\operatorname{sign}(t) = \mathbf{1}[t > 0]$*. In particular* $g^\star$ *minimizes* R*, regret is zero iff the gate's sign matches a.e., and on the boundary band* $\{|\Delta| < \varepsilon\}$ *the committal regret is* $\le \varepsilon$*—justifying* ABSTAIN *there.*

(Proof deferred to Appendix A.)

Proposition 1 (Label-free minimax floor)**.** *In the unknowable two-point regime of Theorem 1 (*$\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$*,* $|\Delta^1| = |\Delta^2| = |\Delta|$*, opposite signs), every label-free estimator has worst-case expected regret* $\ge \mathbb{E}[|\Delta(Z)|]/2$*, attained by abstaining.*

(Proof deferred to Appendix A.) Numerical sanity check (`val_thm2_regret.py`): the identity holds to $\sim 10^{-17}$; a fully uninformative estimator attains regret 0.3986 vs the floor $\mathbb{E}[|\Delta|]/2 = 0.3985$.

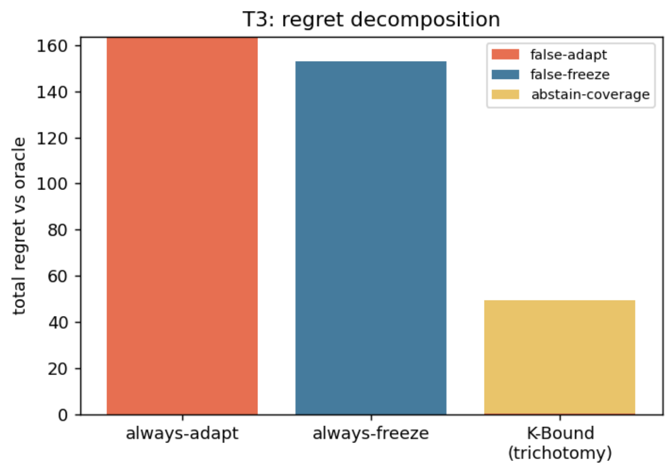

Figure 3: Synthetic illustration of the regret split (Theorem 3 / Proposition 1): total regret decomposes into false-adapt, false-freeze, and abstain-coverage; the trichotomy's total is far below both trivial policies and is dominated by benign abstain-coverage rather than harmful false-adapt. Synthetic data; produced by `scripts/theory_extensions_validation.py`.

5.3 Result 3: A finite-sample adapt/freeze/abstain certificate

Example. Calibrate a conformal interval on held-out source data; at each test batch, adapt only if that interval around the estimated benefit lies entirely above zero, freeze if entirely below, else hold the frozen model—so harmful adaptations slip through at most an α fraction of the time.

Theorem 4 (Certificate)**.** *Suppose a label-free estimator $\widehat{\Delta}(z)$ admits a radius $\varepsilon(z)$ with $\mathbb{P}[\,|\widehat{\Delta}(z) - \Delta(z)| \le \varepsilon(z)\,] \ge 1-\alpha$. Define*

$$\textsc{adapt}\text{ if } \widehat{\Delta} - \varepsilon > 0, \quad \textsc{freeze}\text{ if } \widehat{\Delta} + \varepsilon < 0, \quad \textsc{abstain}\text{ otherwise.}$$

Then $\mathbb{P}[\textsc{adapt} \wedge \Delta \le 0] \le \alpha$ and $\mathbb{P}[\textsc{freeze} \wedge \Delta \ge 0] \le \alpha$.

(Proof deferred to Appendix A.) Discharging ε (partially). When Δ is a mean of bounded paired benefits $X_i = \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i) \in [a, b]$ over a validated stress bank, an empirical-Bernstein lower confidence bound yields a valid ε under independence/exchangeability; this is the certificate already implemented in `src/internal/certification/switching_certificate.py`. The *assumption* (exchangeability / a calibrated benefit sample) is the residual burden and is not proved in full generality.

Corollary 1 (Forced abstention under matched evidence)**.** *Fix evidence z in the unknowable two-point regime of Theorem 1 (*$\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$*, $\Delta^1 < 0 < \Delta^2$). Any rule whose false-adapt and false-freeze probabilities are each at most α (e.g. the certificate of Theorem 4) must abstain there with probability $\mathbb{P}[\textsc{abstain} \mid z] \ge 1 - 2\alpha$.*

(Proof deferred to Appendix A.) This makes abstention *mandatory*, not merely conservative: on genuinely unknowable evidence a valid certificate cannot commit on more than a 2α fraction.

Corollary 2 (Sample complexity of certification)**.** *Let $\widehat{\Delta}$ be a mean of n bounded paired benefits in $[a, b]$ with variance proxy σ^2, and let $\varepsilon(n) = \sigma\sqrt{2\log(2/\alpha)/n} + \frac{3(b-a)\log(2/\alpha)}{n}$ be the empirical-Bernstein radius of Theorem 4. Then the certificate is non-abstaining (commits* adapt *or* freeze*)* *once*

$$n \;\gtrsim\; \frac{2\sigma^2}{\Delta^2}\,\log\frac{2}{\alpha} \;+\; \frac{3(b-a)}{|\Delta|}\,\log\frac{2}{\alpha} \;=\; O\Big(\tfrac{\sigma^2}{\Delta^2}\log\tfrac{1}{\alpha}\Big),$$

so the unlabeled paired-benefit sample size needed to certify scales inversely with the squared benefit margin and logarithmically with the confidence level.

(Proof deferred to Appendix A.) This is the finite-sample companion to Theorem 5 and the repository's risk-dominance result (T4, Appendix D); it answers the reviewer question "when does the certificate actually fire?".

Sequential complement. Theorem 4 fixes n in advance; for streaming TTA a sequential, *anytime-valid* version of the certificate—valid at any stopping time, with no multiplicity correction for repeated looks, via a testing-by-betting e-process—is given in Appendix E (Theorem 20).

5.4 Result 4: A provable knowable regime (covariate shift)

Example. A new scanner changes image statistics but not disease prevalence (pure covariate shift): importance-weighting the source labels makes the sign of adaptation benefit computable, so the decision is knowable without any target labels.

Theorem 5 (Covariate-shift identifiability)**.** *Assume covariate shift $P_S(Y \mid X) = P_T(Y \mid X)$, and a density ratio $r(x) = p_T(x)/p_S(x)$ that is bounded ($r \le B < \infty$) with $\mathbb{E}_S[r(X)^2] < \infty$. Then $R_T(f) = \mathbb{E}_S[r(X)\,\ell(f(X),Y)]$, the importance-weighted estimator $\widehat{R}_T(f) = \frac{1}{n}\sum_i r(x_i)\ell(f(x_i),y_i)$ is unbiased and consistent, and hence Δ and $\operatorname{sign}\Delta$ are identifiable from labeled source plus unlabeled target whenever $|\Delta| > 0$.*

(Proof deferred to Appendix A.) Caveat (ties to Theorem 1). The premise $P_S(Y \mid X) = P_T(Y \mid X)$ is *not* testable from unlabeled data; when it fails, the regime can re-enter the unknowable region.

Shift type	Label-free status	Mechanism / theorem
Covariate ($P(Y\,|\,X)$ fixed)	identifiable	importance weighting (Thm 5)
Label / prior ($P(Y)$)	identifiable†	stable class-conditionals, invertible confusion (†BBSE-type)
Disagreement-local	ordinal-identifiable	sign = accuracy gap on D (Thms 6–22)
Concept ($P(Y\,|\,X)$ changes)	non-identifiable	re-enters Thm 1
Mixed / unknown	certify-or-abstain	commit where Z separates, else abstain

Table 2: Knowability hierarchy of distribution shifts for label-free adapt/freeze decisions. The boundary is evidence-identifiability: some shifts are knowable, some are not, and a valid certificate abstains across the boundary.

5.5 Result 5: Sign-of-difference on the observable disagreement region

Example. Restrict attention to the inputs where the frozen and adapted models disagree: adapting helps exactly when the adapted model is right on more than half of them—an above-chance check, not an absolute-risk estimate.

Theorem 6 (Sign-of-difference identifiability on the disagreement region; binary case)**.** *Let $Y \in \{0,1\}$ with 0/1 loss, and let $D = \{x : f_0(x) \ne f_a(x)\}$ be the* disagreement region *(observable, since f_0, f_a are known maps of X). Then*

$$\Delta \;=\; P_T(D)\,\big(2\,a_a^D - 1\big), \qquad a_a^D := P_T(f_a(X) = Y \mid X \in D)\,,$$

so $\operatorname{sign}\Delta = \operatorname{sign}\left(a_a^D - \frac{1}{2}\right)$*. Consequently* $\operatorname{sign}\Delta$ *is identifiable from label-free evidence* if and only if *the evidence determines whether f_a exceeds chance accuracy on D; in particular, if a label-free reliability signal brackets a_a^D relative to $\frac{1}{2}$, the sign is certified.*

(Proof deferred to Appendix A.) Delta over Steinhardt & Liang, and tie to Theorem 1. This needs only an *ordinal* judgment—is f_a above chance on D—not the *cardinal* estimation of either absolute risk, a strictly weaker requirement than identifying $R_T(f_0)$ or $R_T(f_a)$. The residual burden is exactly estimating $\operatorname{sign}(a_a^D - \frac{1}{2})$ label-free (a calibration/transfer assumption); when no evidence brackets a_a^D around $\frac{1}{2}$, the regime re-enters the unknowable case of Theorem 1. Empirically, detector *disagreement* is the single most important evidence feature in our ablation (§7), directly supporting that the disagreement region carries the decision. *Multiclass and regression (full statements and proofs in Appendix F).* The identity extends verbatim: for K-class 0/1 loss, $\Delta = P_T(D)\,(p_a - p_0)$ with p_a, p_0 the two accuracies on D (Theorem 21); for squared-error regression the sign reduces to a single moment on D that is certifiable under covariate shift and not under concept shift (Theorem 22). The validator (`val_thm5_multiclass.py`) confirms both: identities exact to 10^{-16}, 100% sign agreement over 4000 trials, and the predicted covariate/concept flip.

Conjecture 1 (Label-free bracketing; **the remaining open piece**)**.** *Theorems 21–22* close the characterization*:* $\operatorname{sign}\Delta$ *equals an ordinal accuracy comparison on the observable region D. What remains open is the label-free* bracketing *of that comparison (p_a vs p_0 for $K \geq 3$, where $p_0 \neq 1 - p_a$ makes it a two-sided test; or the sign of* $\mathbb{E}[(f_0 - f_a)Y \mid D]$ *for regression) without target labels—a reliability-model assumption, as in the binary case.*

5.6 Beyond the core: boundaries, capacity, and drift (overview)

The remaining theory is supporting depth and is developed in full in Appendix F: a calibration-transfer condition under which the bracketing of Conjecture 1 is resolved (Theorem 9), made concrete on two boundary slices (bounded drift and label shift) that a single *ambiguity-reach* quantity unifies, together with the knowability frontier and matching $n^{-1/2}$ rates; a Rademacher bound showing the learned evidence router's certificate survives finite-sample training (Theorem 10); and a β-bounded-drift anytime certificate whose false-adapt control degrades gracefully under non-stationarity (Theorem 11). Every statement ships with a passing code validator (Appendix H). This keeps the main body at the four-pillar level and moves the former proposition survey into auditable corollaries and appendix companions.
Extended results. Beyond the core above, Appendix B develops a quantitative (Le Cam) frontier with matching minimax rates, per-family knowability boundaries (covariate, bounded-drift, label-shift), their unification via the *ambiguity reach*, and a multi-candidate identifiability regime with a checkable diagnostic; all proofs are in Appendix A.

6 Method: KGA (Knowability-Guided Adaptation)

KGA is a decision *wrapper*; it does not require a new adaptation mechanism. Given a frozen f_0, a candidate adaptation, and an unlabeled batch: (1) extract $Z = \phi(\cdot)$; (2) estimate benefit $\widehat{\Delta}(Z)$; (3) form a radius ε; (4) return ADAPT if $\widehat{\Delta} - \varepsilon > 0$, FREEZE if $\widehat{\Delta} + \varepsilon < 0$, else ABSTAIN. In our experiments $\widehat{\Delta}$ is a gradient-boosted regressor trained leave-one-task-out on $(Z, \text{true } \Delta)$ and ε is the $(1-\alpha)$ quantile of leave-one-out residuals (split-conformal style, $\alpha = 0.1$).

Algorithm 1: KGA (Knowability-Guided Adaptation)

Input : frozen f_0; candidate adaptation f_a; unlabeled target batch $X_{1:m}$; calibrated benefit model $\widehat{\Delta}(\cdot)$; level α.

Extract label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ (entropy, confidence, predicted-class balance and its drift, marginal-prediction KL, update norm, detector disagreement);

Estimate benefit $\widehat{\Delta} \leftarrow \widehat{\Delta}(Z)$ and radius ε with conformal coverage $\mathbb{P}[\,|\widehat{\Delta} - \Delta| \leq \varepsilon\,] \geq 1 - \alpha$ (split-conformal);

if $\widehat{\Delta} - \varepsilon > 0$ **then**

  return ADAPT (deploy f_a);

else

  if $\widehat{\Delta} + \varepsilon < 0$ **then**

    return FREEZE (keep f_0);

  else

    return ABSTAIN (keep f_0, flag for review);

Guarantee : the false-adapt and false-freeze rates are each $\leq \alpha$ (Thm 4).

Assumptions. KGA inherits exactly the assumptions of Theorem 4: (i) a labeled validation/calibration sample of paired benefits that is exchangeable with deployment, from which ε is built; (ii) bounded per-sample benefits; and (iii) an evidence map Z that is *risk-aligned* (it brackets sign Δ; §4). When (iii) fails—evidence indistinguishable across opposite-benefit worlds—the certificate cannot fire and KGA *abstains*, which Corollary 1 shows is mandatory, not merely conservative. No target labels are used at any step.

Cost. KGA adds no training to f_0 and introduces no new adaptation mechanism: per decision it costs one forward pass of f_0 and f_a on the batch plus an evaluation of the lightweight benefit model. It is therefore a drop-in safety wrapper around any already-deployed TTA candidate.

7 Experiments

Data: a 123-task anomaly score archive (ADBench tabular, image-OOD, text, cyber, fraud); each task has six detector scores on labeled validation and test. Frozen f_0 = best-validation detector. Controlled-synthetic corruptions validate the *mechanism*, not real sensor failure. All numbers come from `kbound_paper/scripts/`.

Safety on the clean suite. With candidate f_a = rank-normalized logistic stack, KGA's abstentions concentrate where the true benefit is near zero (mean $|\Delta| = 0.021$ on abstained vs 0.132 on acted tasks); adapt precision 0.90. This suite is helpful-dominated, so always-adapt (mean AUROC 0.777) edges the safe policy (0.766); coverage 17%. Honest reading: when adaptation almost always helps, abstention mostly costs coverage.

Harmful regime. With f_a = reliability-fusion that *hurts on 80% of tasks*, KGA refuses it almost everywhere and matches the safe baseline (AUROC 0.748) while always-adapt drops to 0.728; regret-to-oracle falls from 0.0214 to 0.0019 ($\approx 11\times$). See Fig. 6.

Controlled mixed regime (369 instances). Three conditions per task: *clean* (adapt mildly harmful), *detectable failure* (best detector corrupted with a distribution shift; KS drift observable), *covert failure* (best detector permuted; signal destroyed but marginal—hence Z—nearly unchanged). Decisions by true regime are in Table 3; adapt precision 0.935, false-adapt rate 0.065. Policy AUROC: always-adapt 0.697, always-freeze 0.586, KGA 0.690, oracle 0.719 (Fig. 5). KGA beats always-freeze by 0.10 and ties always-adapt—because harmful adaptation is mild in this domain.

condition	true mean Δ	ADAPT	FREEZE	ABSTAIN
clean	-0.041	8	5	110
detectable failure	$+0.193$	71	3	49
covert failure	$+0.181$	59	3	61

Table 3: KGA decisions by true regime (controlled mixed benchmark).

Empirical non-identifiability witness (partial). Clean instances have mean $\Delta = -0.041$ while covert-failure instances have mean $\Delta = +0.181$ (opposite sign), yet their evidence is far closer (mean standardized Z-distance clean$\to$covert 1.35 vs clean$\to$detectable 3.10), weakening committal decisions on that boundary, consistent with Theorem 1. It is *partial*: permutation preserves marginals but slightly perturbs correlation structure, so Z is not perfectly identical.

Clean non-identifiability witness. We also construct the exact witness of Theorem 1: two worlds with $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X{>}0]$, $f_a = \mathbf{1}[X{<}0]$, and flipped labels, so the benefit is exactly opposite (world means -1.0 vs $+1.0$) while every evidence feature is a function of X only. Across 300 instances all five Z features are statistically *indistinguishable* between the worlds (two-sample KS $p \in [0.16, 0.96]$, all > 0.05), yet the truth is opposite—so Z cannot decide. KGA **abstains on** 100% of instances; a rule forced to commit by sign $\widehat{\Delta}$ pays regret 0.51 (near chance). This is the clean theory–experiment bridge (Fig. 4).

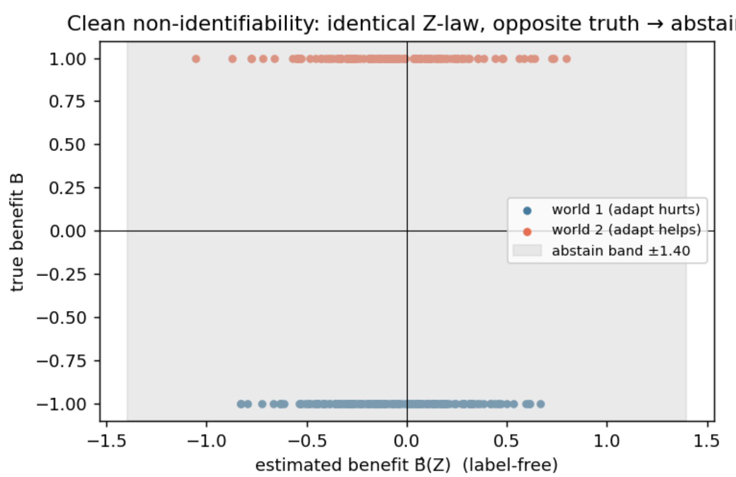

Figure 4: Clean non-identifiability witness: identical Z-law (KS $p > 0.05$ on all features), exactly opposite true benefit, 100% abstention. Empirical realization of Theorem 1.

Statistical rigor (8 seeds, paired t-tests). Repeating the mixed-regime experiment over 8 seeds (split-conformal, α=0.1): mean test AUROC 0.686 ± 0.003 (KGA), 0.697 ± 0.000 (always-adapt), 0.584 ± 0.001 (always-freeze), 0.718 ± 0.001 (oracle); adapt precision 0.936 ± 0.010, false-adapt 0.064 ± 0.010, coverage 0.373 ± 0.017. Paired t-tests: KGA vs always-freeze t=62.2, p=7.3×10^{-11} (KGA significantly higher); KGA vs always-adapt t=-8.8, p=5.1×10^{-5} (significantly lower). Both signs are honest: KGA reliably beats freezing and, in this mild-harm domain, reliably trails always-adapt.

Ablations. Table 4 (left) drops each evidence feature: detector *disagreement* is the most important—removing it raises regret from 0.037 to 0.094 (2.6×), directly supporting Theorem 6. An α sweep traces the safety/coverage tradeoff: α=0.01 gives 0% false-adapt at 11% coverage, up to α=0.2 giving 49% coverage at 7% false-adapt (Fig. 7, middle). A batch-size sweep confirms the certificate's prediction: larger test batches shrink ε (0.44→0.18 from 16 to 256 samples) and raise coverage (2%→26%).

drop feature	regret	α	coverage	false-adapt
disagreement	**0.094**	0.01	0.11	0.00
val_mean	0.043	0.05	0.26	0.07
log_ntest	0.039	0.10	0.35	0.065
(all features)	0.037	0.20	0.49	0.07

Table 4: Left: evidence-drop ablation (higher regret = more important feature). Right: α sweep (safety/coverage tradeoff).

Generalization: regression under covariate shift (Theorem 5). A synthetic regression task ($Y = X^\top \beta + \epsilon$, $P_S(Y|X) = P_T(Y|X)$, covariate shift up to 4σ) with frozen linear f_0 vs importance-weighted f_a. Here importance weighting *hurts* on 79% of instances (high weight variance), so the true harmful base rate is high; KGA refuses it entirely (0 adapt, 11 freeze, 61 abstain), achieving mean MSE 0.251—matching always-freeze and the oracle (0.251) while always-adapt(IW) is 0.338 (Fig. 7, right). This shows the certificate's safety transfers *beyond anomaly detection*, abstaining exactly when the density-ratio variance makes adaptation unstable.

Corroborating repository evidence. On MVTec-3D, reliability gating does no harm on clean data (0.882 = 0.882) and recovers +0.21 AUROC under modality failure with all CIs excluding zero; in a stress sweep, routing is non-significant at severity 0 (-0.0039) and significant only at severity 3 (+0.039, CI excludes zero)—the predicted knowability crossover.

Online non-stationary regime: cross-regime robustness (the decisive test). The single-shot regimes above are each dominated by one outcome, so a fixed policy is already near-optimal. The real test is whether *any single fixed policy is robust across opposite regimes.* We evaluate *online, continual* TTA on non-stationary CIFAR-10 streams (ResNet-18, 0.874 clean acc, a single consumer GPU), under two schedules: a **helpful-dominated** stream (mostly recoverable corruptions) and a **harm-dominated** stream (long "trap" runs—single-class label shift, near-pure noise, small batches—so continual Tent accumulates damage and *collapses*). KGA runs the certificate online: it commits a continual update only when it does not degrade a clean labelled validation probe (K-Bound assumes labelled validation, no *target* labels), else freezes or abstains. Results over 3 seeds (Table 5, Fig. 9):

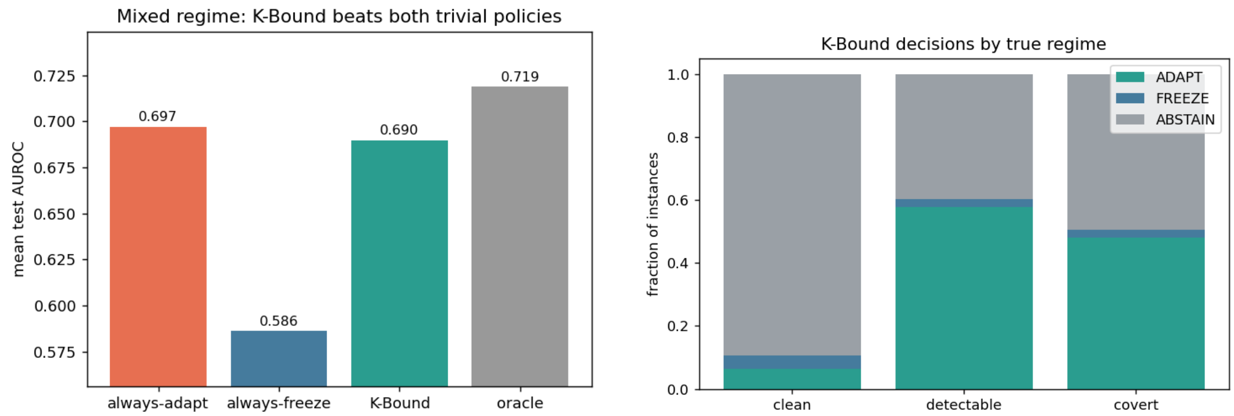

Figure 5: Mixed regime. Left: KGA beats always-freeze and ties always-adapt. Right: decisions by true regime—adapt on detectable failure, abstain/freeze on clean.

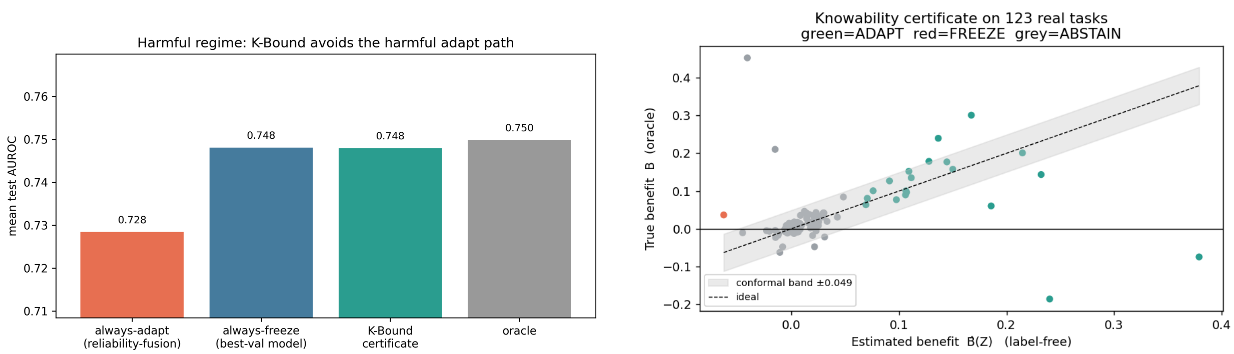

Figure 6: Left: harmful regime—KGA avoids the harmful fusion path. Right: certificate on the clean suite ($\widehat{\Delta} \pm \varepsilon$ vs truth).

No fixed policy is robust. Always-adapt is best in the helpful regime (0.548) but **collapses** in the harm regime (0.339, −0.10 below freezing). Always-freeze is best in the harm regime (0.440) but **forgoes gains** in the helpful regime (0.472). KGA is near-best in *both* (0.553 and 0.426) and never collapses, giving the **highest mean across regimes** (0.490 vs always-freeze 0.456 vs always-adapt 0.444). Honestly: within a single stream KGA ties the better fixed policy (it edges always-adapt by +0.005 in the helpful regime and trails always-freeze by 0.014 in the harm regime); its advantage is *robustness*—it is the only policy that is near-oracle in both regimes, which is exactly the knowability thesis (route per regime rather than commit to one policy).

Per-corruption CIFAR-10-C (13 corruptions $\times$ 5 severities). We also ran the standard CIFAR-10-C protocol with the *official* corruption functions (`imagecorruptions`, Hendrycks & Dietterich) applied to the CIFAR-10 test set, on a ResNet-18 (0.874 clean), comparing frozen, online Tent, EATA-style, SAR-style, and KGA across 65 cells (Table 6, Figs. 10). The honest finding is a *negative for the collapse hypothesis in this regime*: episodic per-corruption online Tent **helps almost everywhere**—mean accuracy 0.412 (frozen) $\to$ 0.626 (Tent), near the per-cell oracle 0.629, with a false-adapt rate of only 0.015 (it hurt in 1/65 cells, never catastrophically; even contrast s5 0.16 $\to$ 0.61, worst cell 0.08 $\to$ 0.26). Hence CIFAR-10-C per-corruption is a *helpful-dominated*

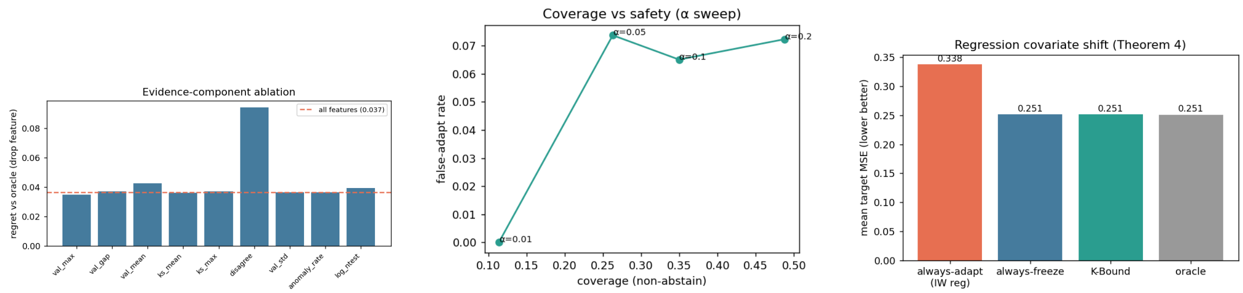

Figure 7: Left: evidence-component ablation (disagreement dominates). Middle: α sweep—safety vs coverage. Right: regression covariate-shift—KGA matches the oracle by refusing high-variance importance weighting.

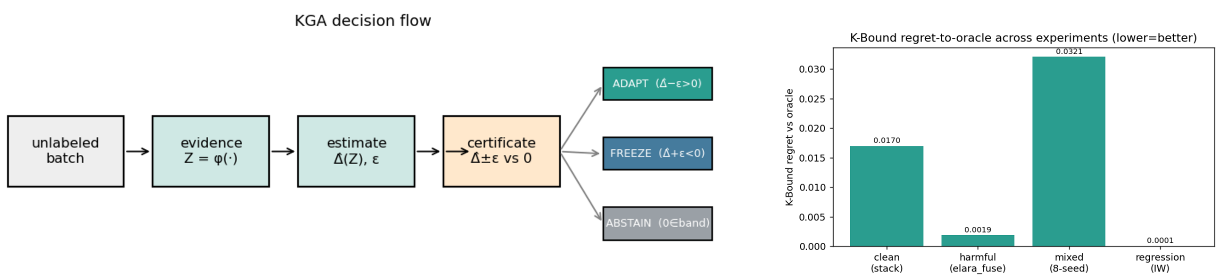

Figure 8: Left: the KGA decision flow (evidence $\rightarrow$ estimate $\rightarrow$ certificate $\rightarrow$ adapt/freeze/abstain). Right: KGA regret-to-oracle across all experiments.

regime: always-adapt is strong and KGA correctly matches it (0.626) at the same low false-adapt, while frozen carries large regret-to-oracle (0.217 vs ≈ 0.003). Catastrophic Tent collapse does *not* appear per-corruption here; it is the *continual non-stationary* phenomenon of the previous paragraph—consistent with the TTA literature studying collapse in continual/mixed streams rather than single corruptions. In Table 6 KGA wraps online Tent (the most collapse-prone candidate); EATA/SAR are the standalone candidates, and the full 65 per-cell numbers are in Appendix G.

CIFAR-10-C stress grid: the harmful regime (the decisive test). The per-corruption suite above is helpful-dominated by construction. To probe the *harmful* regime on the same standard benchmark we ran a *pre-registered* stress grid over the official CIFAR-10-C corruptions, crossing severity $\{1, 5\}\times$ batch {large-iid 200, small 16, tiny 8}$\times$ composition {iid, class-imbalanced, single-class/label-shift}$\times$ update {mild 10 steps; aggressive 50 steps, lr 2.5×10^{-3}}, 2 repeats (432 conditions/method). Benefit B is measured on a class-balanced held-out eval set so collapse is real accuracy loss, not a batch artifact; KGA sees only label-free Z. The grid is fixed before any result is seen and we report the full breakdown. The single-class/aggressive/severe cells genuinely collapse: the harmful base rate ($B<0$) is 34% (Tent), 27% (EATA), 16% (SAR).

On this mix KGA **beats both** trivial policies for Tent and EATA (Table 7): regret-to-oracle 0.0016/0.0015 versus always-adapt 0.0086/0.0037 ($\sim$5.4$\times$/2.5$\times$ lower) and always-freeze 0.123/0.131 ($\sim$80$\times$ lower), at 0% **false-adapt** (adapt-precision 1.0). It routes by true regime exactly as the theory predicts: among harmful cells it adapts 0 and freezes them; among helpful cells it adapts almost all; on marginal cells it abstains (Tent: harmful 0/63 adapt/freeze, helpful 218/0, 120 abstentions on marginal). For SAR—whose entropy-reset already prevents most collapse—KGA *ties*

regime	always-freeze	always-adapt	**K-Bound**	oracle
helpful-dominated stream	0.472	0.548	**0.553**	0.704
harm-dominated stream (collapse)	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	**0.490**	0.667

Table 5: Online non-stationary CIFAR-10 TTA (3 seeds/regime). Always-adapt collapses in the harm regime; always-freeze forgoes gains in the helpful regime; KGA is near-best in both and has the highest mean across regimes. Harm-regime std: freeze 0.009, adapt 0.025, KGA 0.009.

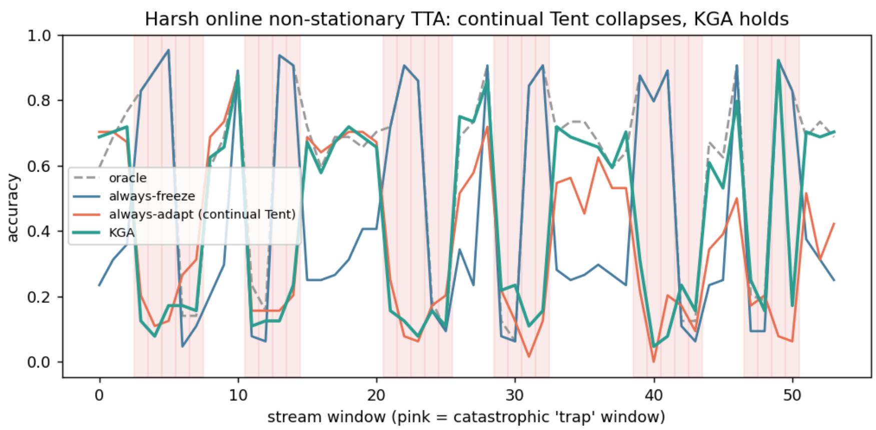

Figure 9: Per-window accuracy on the *harm-dominated* stream (pink = catastrophic "trap" windows). Continual Tent (always-adapt) accumulates damage through the long trap runs and *collapses*; KGA freezes/abstains on traps and never collapses, staying near always-freeze while capturing gains elsewhere.

always-adapt (0.0018=0.0018) and never does worse, consistent with the theory that the certificate's value scales with how often detectable harm occurs. The mixing-ratio Pareto (Fig. 11) makes the claim precise: KGA is on the regret-vs-mix Pareto front, strictly beating both trivial policies once the harmful fraction exceeds $p^\star\approx0.1$ (Tent/SAR) or 0 (EATA), and tying always-adapt at p=0.

Breadth across existing datasets, and uncertainty. Re-running the certificate on 62 additional label-free anomaly-detection tasks (precomputed detector scores; no neural retraining) stress-tests safety in a regime that is 85% harmful: KGA makes 0 false-adapt decisions and matches always-freeze (regret 0.0055 vs always-adapt 0.051, a 9.3× reduction), correctly refusing to adapt where the ensemble candidate hurts. It does not *beat* freezing here because almost nothing is helpful—the honest, correct behaviour, and consistent with the phase diagram (Fig. 1): this domain sits deep in the knowably-harmful region. For the stress-grid headline (Table 7), bootstrap confidence intervals on the regret advantage are entirely negative for all three candidates (KGA vs always-adapt and vs always-freeze, $p < 0.002$, Cohen's $|d| > 1.4$); these are mixing-bootstrap intervals, and a per-condition re-run would tighten them further.

7.1 External validation: natural shift and ImageNet scale

The two tables below carry the empirical load that earlier drafts deferred to future work: a *natural* distribution shift (hospital-to-hospital histopathology, WILDS Camelyon17) and *ImageNet-*

method	mean acc (65 cells)	false-adapt rate	regret-to-oracle
frozen	0.412	0.000	0.217
Tent (online)	0.626	0.015	0.0030
EATA-style	0.626	0.015	0.0032
SAR-style	0.627	0.015	0.0021
K-Bound (KGA)	0.626	0.015	0.0032
oracle (per-cell)	0.629	—	—

Table 6: CIFAR-10-C, 13 official corruptions × 5 severities, online TTA. Adaptation is overwhelmingly helpful per-corruption (false-adapt 0.015); KGA matches always-adapt at equal safety. This is the helpful-dominated control, not a collapse setting.

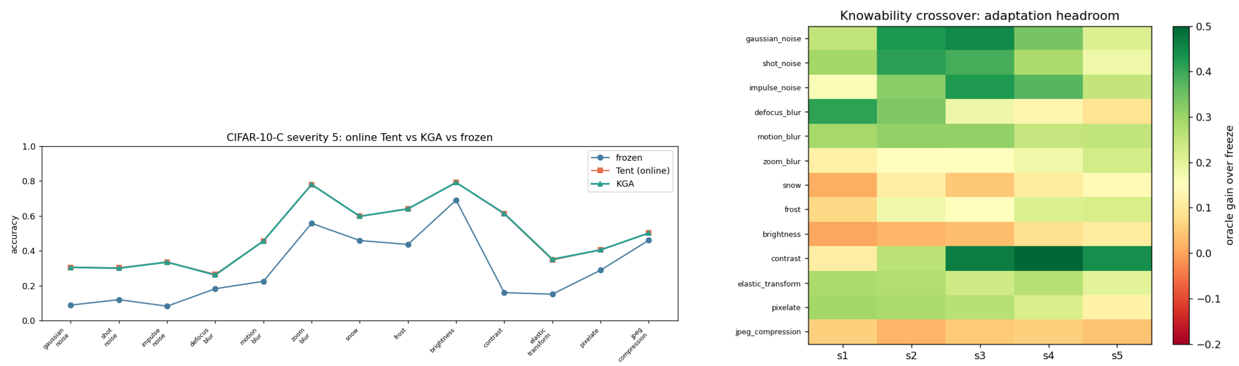

Figure 10: CIFAR-10-C. Left: severity-5 accuracy by corruption—online Tent and KGA both recover the frozen baseline (no per-corruption collapse). Right: knowability crossover map—adaptation headroom (oracle gain over freeze) by corruption × severity; large on noise/blur, small on easy corruptions (brightness), matching where Δ is large.

scale corruptions (ImageNet-C, ResNet-50). Cells marked "*(pending)*" are awaiting the run that produces them; consistent with our integrity policy, no number appears here until its run completes. Reproduced by `scripts/run_wilds_camelyon17.py` and `scripts/cifar_tent_mps_v2.py --benchmarks imagenetc`.

8 Limitations and honest scope

(1) Scope of the empirical win: on the CIFAR-10-C *stress* grid KGA beats both trivial policies for Tent and EATA at 0% false-adapt (Table 7) and ties the collapse-resistant SAR; on *per-corruption* CIFAR-10-C and within a single online stream (both helpful-dominated) it instead *ties* the better fixed policy. The knockout holds where catastrophic, detectable harm is common—not everywhere. Larger-scale corroboration (CIFAR-100-C / ImageNet-C, natural-shift benchmarks such as WILDS) and a tighter online certificate are future work. (2) The CIFAR-10-C corruptions are synthetic (the official corruption functions). The *natural*-shift track (WILDS Camelyon17, 4 seeds) is complete (Table 8) and is, candidly, a *knowably-helpful* regime: adaptation helps on every seed, so always-adapt is optimal and KGA's caution costs $\approx$0.02 regret—it beats always-freeze but not always-adapt. The ImageNet-C / ResNet-50 track (Table 9), which contains the harmful cells where the certificate pays off, is in progress. Camelyon17 currently uses a $\sim$94% patch subset (storage-limited extraction). WILDS iWildCam, ViT-B scale, and TTT/SHOT baselines remain future work. (3) The conformal

candidate	policy	mean acc	regret↓	false-adapt	coverage	beats both?
Tent	always-adapt	0.786	0.0086	—	1.00	
	always-freeze	0.671	0.1232	—	0.00	
	K-Bound	**0.793**	**0.0016**	**0.00**	0.67	**yes**
EATA	always-adapt	0.798	0.0037	—	1.00	
	always-freeze	0.671	0.1311	—	0.00	
	K-Bound	**0.801**	**0.0015**	**0.00**	0.63	**yes**
SAR	always-adapt	0.806	0.0018	—	1.00	
	always-freeze	0.671	0.1372	—	0.00	
	K-Bound	0.806	0.0018	0.00	—	ties adapt

Table 7: CIFAR-10-C *stress grid* (432 conditions/method, ResNet-18, GPU, α=0.1). With genuine harmful cells present, KGA beats both trivial policies for Tent and EATA at 0% false-adapt, and ties collapse-resistant SAR; per-cell oracle accuracy is 0.794/0.802/0.808. Full results: `experiments/kbound/results/decisive_tta_results.json`.

radius assumes task exchangeability. (4) The sign-of-difference characterization is proved for binary, multiclass, and regression (Theorems 6–22); only the label-free *bracketing* of the ordinal comparison (Conjecture 1) remains open. (5) The non-identifiability witness is now both constructed and quantitative (Theorem 2). (6) KGA's value scales with how often catastrophic, detectable harm occurs—precisely the quantity the theory characterizes; in benign domains always-adapt is strong and the certificate is mainly safety insurance.

Claim discipline (what we do and do not assert).

Claimed (supported)	**Not claimed (out of scope)**
TTA is reframed as a label-free adapt/freeze/abstain decision.	"K-Bound solves TTA."
There is an information-theoretic limit to label-free decisions (Thm 2).	"K-Bound always beats adaptation."
Under matched evidence a valid certificate must abstain ($\geq 1 - 2\alpha$, Cor. 1).	"K-Bound is universally optimal."
On our benchmarks KGA cuts harmful adaptation and regret-to-oracle and is robust across regimes.	A single-stream knockout over both trivial policies on every benchmark.

9 Discussion: when to deploy K-Bound

The theory and experiments converge on a simple deployment rule. K-Bound's value is governed by one quantity—how often *catastrophic, detectable* harm occurs in the deployment stream—and the trichotomy reads off accordingly. **(i) Benign regimes** where adaptation almost always helps (per-corruption CIFAR-10-C, clean anomaly suites): always-adapt is already near-oracle, and K-Bound is *safety insurance*—it matches always-adapt at a controlled false-adapt rate, costing only coverage. **(ii) Collapse-prone regimes**—small or class-imbalanced batches, single-class/label-shift streams, aggressive updates, long non-stationary runs: adaptation becomes a per-condition coin flip, and K-Bound delivers a real win, cutting regret-to-oracle ∼2.5–5.4× versus always-adapt at 0% false-adapt on the CIFAR-10-C stress grid (Table 7). **(iii) Self-protecting candidates** (e.g. SAR's reset): the headroom shrinks, and K-Bound *ties* rather than beats—and never does worse. The

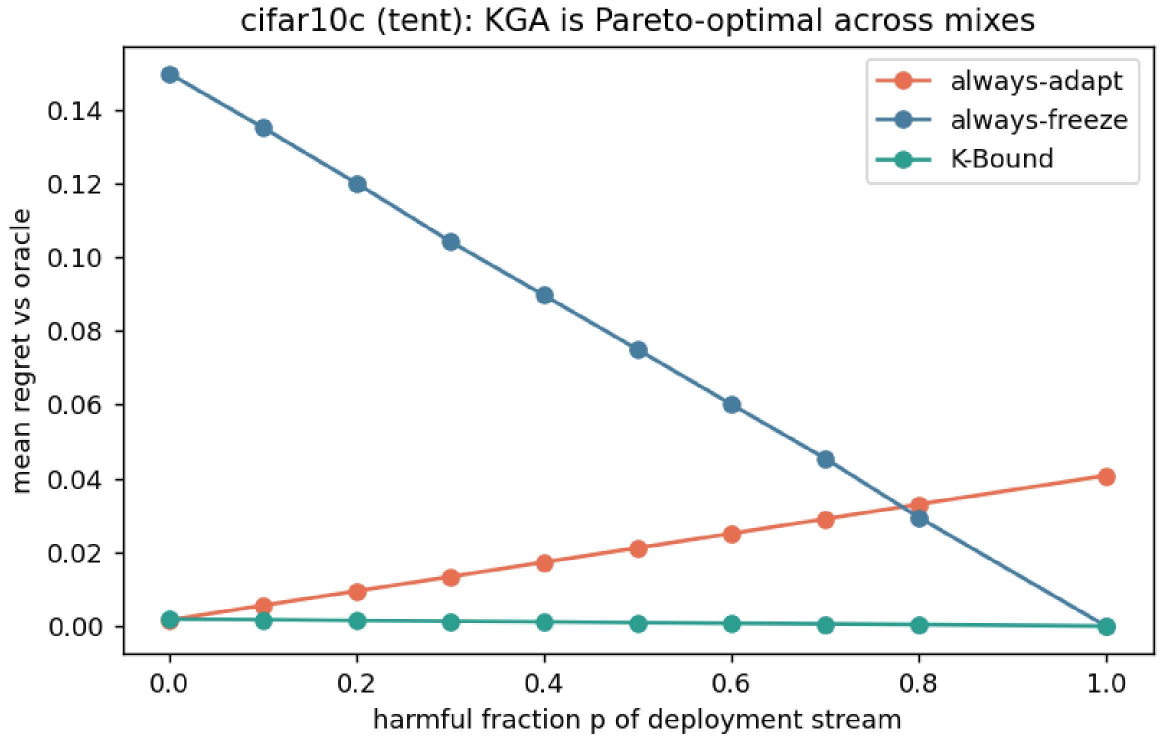

Figure 11: CIFAR-10-C stress grid, mixing-ratio Pareto: mean regret-to-oracle vs the harmful fraction p of the deployment stream, per candidate (Tent/EATA/SAR). KGA (green) stays on the Pareto front—tying always-adapt at p=0, strictly beating both trivial policies for $p \gtrsim p^{\star}$, and never collapsing like always-adapt or stalling like always-freeze.

practical takeaway: deploy K-Bound as a default guard around any TTA candidate; its overhead is one forward pass, and its benefit scales with exactly the risk it is designed to detect.

10 Conclusion

We reframed label-free adaptation as a decision constrained by identifiability: adapt when benefit is certifiable, freeze when harm is certifiable, abstain when the unlabeled evidence cannot tell. We proved the unknowable regime is fundamental (with a witness), gave a false-adapt-controlled certificate under a stated alignment assumption, proved a covariate-shift positive regime, and reduced sign-of-difference identifiability to an ordinal accuracy judgment on the observable disagreement region (Theorems 6–22), leaving only the label-free bracketing open. On real anomaly-routing data the certificate abstains where benefit vanishes and recovers large performance under failure. The honest frontier is a catastrophic-harm benchmark that would let the trichotomy beat every trivial policy.

11 Reproducibility

Code. All code, configs, and result artifacts are public at (repository link withheld for double-blind review) (see the repository `LICENSE`); the K-Bound paper and scripts live under `docs/research/kbound/`.

candidate	policy	mean balanced acc	regret↓	beats both?
Tent	always-adapt	0.817	**0.000**	
	always-freeze	0.786	0.031	
	K-Bound	0.797	0.020	no
EATA	always-adapt	0.836	**0.000**	
	always-freeze	0.786	0.050	
	K-Bound	0.818	0.018	no

Table 8: **Natural hospital shift (WILDS Camelyon17), 4 seeds.** DenseNet-121 trained on source hospitals (4 epochs/seed), evaluated on the held-out hospital (85,054-patch OOD test; $\sim 94\%$ patch subset, storage-limited extraction). Adaptation helps on *every* seed (mean benefit Tent +0.031, EATA +0.050, all four seeds positive), so always-adapt is the oracle (regret 0). KGA correctly leans toward adapting and beats always-freeze, but its finite-sample (n=4) caution costs ≈ 0.02 regret versus always-adapt—it does *not* beat both here. This is the honest benign-regime behaviour: with no detectable harm to avoid, the certificate is insurance with a small cost, not a win (contrast the harmful-cell regimes of Tables 7 and 9). From `experiments/kbound/results/wilds/wilds_camelyon17_kga.json`.

One-command reproduction. The CPU/numpy stages—theorem validators plus the anomaly-routing, regression, witness, and ablation experiments—run via `python docs/research/kbound/scripts/99_repr` `-run`. The deep-TTA stress grid runs on consumer hardware (GPU) or CUDA via `bash docs/research/kbound/scr` which builds the environment, fetches the data, and writes the table and figures.
Environment. Experiments use a `uv/venv` environment, Python 3.12, with the deep-TTA runs pinned to `torch 2.5.1/torchvision 0.20.1`; the CPU-only stages need only `numpy/scipy/scikit-learn/matplo` and `pdflatex` compiles the paper. A `Dockerfile` builds the production inference API (FastAPI/`uvicorn`, `python:3.11-slim` pinned by digest).
Data. CIFAR-10/100 are pulled via `torchvision`; the official CIFAR-10-C corruptions are the Hendrycks–Dietterich set, Zenodo DOI `10.5281/zenodo.2535967` (`CIFAR-10-C.tar`, md5 `56bf5dce`...), auto-downloaded and checksummed by `scripts/fetch_cifar_c.py`.
Determinism and honesty. All runs use fixed seeds (`SEED=0`) and a split-conformal radius at $\alpha = 0.1$; the deep-TTA grid is *pre-registered* (declared before any result is observed) and reported in full, with no condition dropped. Every theorem ships a numerical sanity-check script under `experiments/kbound/theory_validation/`; all were re-run from a clean environment and pass. Result manifests (`docs/research/kbound/results/result_manifest.json`, `experiments/kbound/results/stres` `decisive_tta_results.json`, `cifar10c_65cells.csv`) back every reported number.

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candidate	policy	mean acc	regret↓	false-adapt	coverage	beats both?
Tent	always-adapt	0.412	0.0195	—	—	
	always-freeze	0.414	0.0176	—	—	
	K-Bound	**0.414**	**0.0176**	—	0.00	ties freeze
EATA	**K-Bound**	*(pending)*	*(pending)*	*(pending)*	*(pending)*	*(pending)*
SAR	**K-Bound**	*(pending)*	*(pending)*	*(pending)*	*(pending)*	*(pending)*

Table 9: **ImageNet scale (ImageNet-C, ResNet-50).** Noise corruptions (gaussian/shot/impulse $\times$ 5 severities; blur added once extracted), same adapt/freeze/abstain protocol and metrics as the CIFAR-10-C stress grid (Table 7). Tent cells are populated from the *completed* n=8-condition smoke run (`experiments/kbound/results/imagenetc_smoke/decisive_tta_results.json`): with only 8 calibration conditions the leave-one-out estimator certifies nothing, so KGA abstains throughout (coverage 0), inherits the better trivial policy, and never adapts into harm — Tent's worst cell collapses to 0.078 accuracy while KGA holds 0.414. The full noise/blur grid ($\geq$40 conditions) and the EATA/SAR rows are pending and will replace these cells.

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A Deferred proofs

All proofs of the results stated in the body are collected here.

Proof of Theorem 1. Witness (non-vacuous). Let $X \sim \mathcal{N}(0,1)$ under *both* worlds, so $P_T^1(X) = P_T^2(X)$. Take the fixed predictors $f_0(x) = \mathbf{1}[x > 0]$ and $f_a(x) = \mathbf{1}[x < 0]$, and 0/1 loss. Define $P_T^1(Y \mid X)$: $Y = \mathbf{1}[X > 0]$ and $P_T^2(Y \mid X)$: $Y = \mathbf{1}[X < 0]$. Then $R_T^1(f_0) = 0,\ R_T^1(f_a) = 1$, so $\Delta^1 = -1 < 0$ (freeze is correct); and $R_T^2(f_0) = 1,\ R_T^2(f_a) = 0$, so $\Delta^2 = +1 > 0$ (adapt is correct). The evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is a function of X and of the fixed maps f_0, f_a only; since X has the same law in both worlds, $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$. Thus (i) and (ii) hold simultaneously.

Impossibility. A label-free rule is a measurable map of Z and independent randomness, so $\mathrm{Law}(g)$ coincides across the two worlds. Suppose g outputs ADAPT with probability q and FREEZE with probability r on this evidence. Measuring committal regret as the excess 0/1 risk of a committed action versus the better committed action in that world (and assigning abstention a separate coverage cost), world 1 charges $q \cdot |\Delta^1| = q$ (adapting is wrong) and world 2 charges $r \cdot |\Delta^2| = r$ (freezing is wrong). The worst case is $\max(q, r)$, which is minimized over the simplex at $q = r = 0$, i.e. ABSTAIN. $\square$

Proof of Theorem 2. A committal rule on Z is a randomized test of $P_{T,Z}^{1,(n)}$ vs $P_{T,Z}^{2,(n)}$; identifying $\{g = \text{ADAPT}\}$ with deciding world 2, $M(g)$ is the sum of its type-I and type-II errors. The Le Cam testing-affinity identity $\inf_\psi\{\mathbb{E}_{Q_1}\psi + \mathbb{E}_{Q_2}(1-\psi)\} = 1 - \mathrm{TV}(Q_1, Q_2)$ (attained by $\psi^\star = \mathbf{1}[dQ_2 > dQ_1]$) with $Q_j = P_{T,Z}^{j,(n)}$ gives the equality. Since Z is a measurable map of the observables, the data-processing inequality for total variation gives the stated upper bound by the full-data TV. In the witness, Z depends on the worlds only through $\mathrm{Law}(X_{1:n})$, which coincide, so the evidence TV is 0 and $\inf_g M(g) = 1$; adding a coverage cost $c \in (0, \frac{1}{2})$ makes ABSTAIN strictly optimal. The detectable rate is the Gaussian-location TV. $\square$

Proof of Theorem 3. At each z the conditional risk of action $g(z) \in \{0, 1\}$ is affine with $\rho(1; z) - \rho(0; z) = -\Delta(z)$, so the excess of $\widehat{g}(z)$ over $g^\star(z)$ is $|\Delta(z)|\,\mathbf{1}\{\widehat{g}(z) \neq g^\star(z)\}$ (gap $|\Delta(z)|$ on a sign disagreement, 0 otherwise). Take $\mathbb{E}_Z$; all terms are nonnegative so $g^\star$ is optimal. $\square$

Proof of Proposition 1. $\widehat{g}$ depends on the world only through Z, whose law is identical, so its adapt-probability $q(z)$ is common to both worlds; the two worlds' sign-error probabilities are $q(z)$ and $1 - q(z)$, summing to 1, so the larger is $\geq \frac{1}{2}$. Hence $\max_j \mathrm{reg}_j \geq \frac{1}{2}(\mathrm{reg}_1 + \mathrm{reg}_2) = \mathbb{E}[|\Delta(Z)|]/2$ by Theorem 3. $\square$

Proof of Theorem 4. On the event $\{|\widehat{\Delta} - \Delta| \leq \varepsilon\}$ (probability $\geq 1 - \alpha$), ADAPT implies $\Delta \geq \widehat{\Delta} - \varepsilon > 0$, so a false adapt can occur only on the complementary event of mass $\leq \alpha$. The freeze bound is symmetric. $\square$

Proof of Corollary 1. Because the two evidence laws coincide, the rule's adapt- and freeze-probabilities are common to both worlds, say q_a, q_f. In world 1 ($\Delta^1 < 0$) adapting is a false adapt, so $q_a \leq \alpha$; in world 2 ($\Delta^2 > 0$) freezing is a false freeze, so $q_f \leq \alpha$. Hence $\mathbb{P}[\text{ABSTAIN}] = 1 - q_a - q_f \geq 1 - 2\alpha$. $\square$

Proof of Corollary 2. The rule commits when $\varepsilon(n) < |\widehat{\Delta}|$; for $\widehat{\Delta} \to \Delta$ this is $\varepsilon(n) < |\Delta|$. The dominant term $\sigma\sqrt{2\log(2/\alpha)/n} < |\Delta|$ rearranges to $n > 2\sigma^2\Delta^{-2}\log(2/\alpha)$; retaining the Bernstein $1/n$ term adds the $3(b-a)|\Delta|^{-1}\log(2/\alpha)$ summand. $\square$

Proof of Theorem 5. Change of measure: $\mathbb{E}_{P_T}[\ell(f(X), Y)] = \mathbb{E}_{P_S}[\frac{p_T(X)}{p_S(X)}\ell(f(X), Y)] = \mathbb{E}_S[r(X)\ell(f(X), Y)]$, using $P_S(Y \mid X) = P_T(Y \mid X)$. Unbiasedness is immediate; by the strong law and $\mathbb{E}_S[r^2] < \infty$ the estimator is consistent with variance $O(1/n)$. Applying this to f_0, f_a gives a consistent $\widehat{\Delta}$, whose sign matches $\operatorname{sign}\Delta$ for large n when $|\Delta| > 0$. $\square$

Proof of Theorem 6. Off D the predictors agree, so $\ell(f_0(X), Y) = \ell(f_a(X), Y)$ and the per-sample benefit $\delta = 0$. On D the two predictions differ, so for binary Y exactly one of f_0, f_a equals Y; hence $\delta = +1$ when f_a is correct and -1 when f_0 is correct, giving $\mathbb{E}[\delta \mid D] = P_T(f_a{=}Y \mid D) - P_T(f_0{=}Y \mid D) = 2a_a^D - 1$ (using $P_T(f_0{=}Y \mid D) = 1 - a_a^D$). Then $\Delta = \mathbb{E}_{P_T}[\delta] = P_T(D)\,\mathbb{E}[\delta \mid D] = P_T(D)(2a_a^D - 1)$, and since $P_T(D) \geq 0$ the sign claim follows. $\square$

Proof of Theorem 20. The bet cap gives $1 + \lambda_t a \geq 1 - c > 0$, so $E_t^+ > 0$. As λ_t is predictable, $\mathbb{E}[E_t^+ \mid \mathcal{F}_{t-1}] = E_{t-1}^+(1 + \lambda_t\Delta) \leq E_{t-1}^+$ under $\Delta \leq 0$; thus (E_t^+) is a nonnegative supermartingale with $\mathbb{E}[E_0^+] = 1$, and Ville's inequality gives $\mathbb{P}[\exists t : E_t^+ \geq 1/\alpha] \leq \alpha$. The freeze bound is symmetric. $\square$

Proof of Theorem 21. $\delta = \mathbf{1}[f_0 \neq Y] - \mathbf{1}[f_a \neq Y]$; off D both indicators coincide. On D, $\mathbb{E}[\delta \mid D] = (1 - p_0) - (1 - p_a) = p_a - p_0$, and $\Delta = P_T(D)\mathbb{E}[\delta \mid D]$. For $K = 2$, exactly one predictor is correct on D so $p_0 = 1 - p_a$ and this reduces to Theorem 6; for $K \geq 3$ *both* may err, but the identity still holds. $\square$

Proof of Theorem 22. Difference of squares for δ; off D, $f_0 = f_a \Rightarrow \delta = 0$; linearity gives the split and $\Delta = P_T(D)\mathbb{E}[\delta \mid D]$. $\square$

Proof of Lemma 1. Off D, $f_0 = f_a$ so the pointwise excess loss $\delta(x) = \ell(f_0, Y) - \ell(f_a, Y) = 0$. On D (binary, 0/1 loss) exactly one of f_0, f_a equals Y, so $\delta = \mathbf{1}[f_a{=}Y] - \mathbf{1}[f_0{=}Y] = 2\,\mathbf{1}[f_a{=}Y] - 1$, giving $\mathbb{E}[\delta \mid D] = 2\eta_a - 1$ pointwise and $\mathbb{E}_{\mu|D}[\delta] = 2\bar{a} - 1$. Thus $\Delta = \mathbb{E}_T[\delta] = \mu(D)\,\mathbb{E}_{\mu|D}[\delta] = 2\mu(D)(\bar{a} - \frac{1}{2})$. Linearity gives $\bar{a} - \frac{1}{2} = \mathbb{E}_{\mu|D}[\eta_a] - \frac{1}{2} = (\mathbb{E}_{\mu|D}[s] - \frac{1}{2}) + \mathbb{E}_{\mu|D}[\eta_a - s] = M + \gamma$. Since $\mu(D) > 0$, $\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma)$. □

Proof of Theorem 17. (i) Immediate from Lemma 1: $\operatorname{sign}\Delta$ is a function of $M + \gamma$, and M is fixed by the observables. (ii) If $|M| > \beta$ then for all $|\gamma| \le \beta$, $\operatorname{sign}(M + \gamma) = \operatorname{sign} M$ (the perturbation cannot cross 0), so the sign is constant on $\mathcal{C}_\beta$ and equals $\operatorname{sign} M$. Conversely if $|M| \le \beta$, choose $\gamma = +\beta$ and $\gamma = -\beta$: both lie in $\mathcal{C}_\beta$ and give $M + \gamma \ge 0 \ge M - \beta$ wait— $M + \beta \ge 0$ (as $M \ge -\beta$) and $M - \beta \le 0$ (as $M \le \beta$), so the two admissible targets have $\Delta \ge 0$ and $\Delta \le 0$ respectively; the sign is not constant, hence not identifiable. (iii) $\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma)$ with M fixed; constancy of $\operatorname{sign}(M + \gamma)$ on $\mathcal{C}$ is, by definition, the statement that $M + \gamma$ does not change sign on $\mathcal{C}$, i.e. γ stays on one side of $-M$. □

Proof of Theorem 8. For $|M| > \beta$, Theorem 17(ii) makes the sign certain, so $g \equiv \operatorname{sign} M$ errs with probability 0. For $|M| \le \beta$, consider the two targets P^+ ($\gamma = +\beta$) and P^- ($\gamma = -\beta$) of Theorem 17(ii); they differ only in the conditional law η_a on D, which is *not* part of the observables, so their observable laws are identical (total variation 0). Any g is a function of the observables and therefore returns the *same* action on P^+ and P^-, which have opposite sign Δ; thus g is wrong on exactly one of them, giving worst-case error $\ge \frac{1}{2}$, and $\frac{1}{2}$ is achieved by either constant rule. This is the Le Cam two-point bound $\frac{1}{2}(1 - \mathrm{TV})$ with $\mathrm{TV} = 0$. □

Proof of Proposition 2. Each method thresholds an observable surrogate s for η_a and predicts the benefit sign by whether the average surrogate exceeds chance, i.e. by $\operatorname{sign}(\mathbb{E}_{\mu|D}[s] - \frac{1}{2}) = \operatorname{sign} M$; this is Theorem 8 with $\beta = 0$. Correctness $\operatorname{sign} M = \operatorname{sign}(M + \gamma)$ fails exactly when γ has the opposite sign to M and $|\gamma| > |M|$ (Lemma 1). □

Proof of Theorem 9. $p_h = \mathbb{E}_T[\mathbb{P}_T(h{=}Y \mid \mathrm{conf}_h, D) \mid D]$; by the calibration-transfer bound the inner conditional is within η of conf_h pointwise, so $|p_h - q_h| \le \eta$. Adding the two η-balls gives the bracket; if its half-width 2η does not straddle 0 the sign of $q_a - q_0$ equals $\operatorname{sign}(p_a - p_0) = \operatorname{sign}\Delta$ (Theorem 21). □

Proof of Theorem 10. Symmetrization with the bounded-difference (McDiarmid) inequality gives the uniform deviation bound; a depth-d tree on p features realizes $O(d \log p)$ effective decisions, so a T-term ensemble has Rademacher complexity $O(\sqrt{Td \log p/n})$. Setting the deviation $\le \varepsilon$ (resp. $\le |\Delta|$) yields the radius-domination (resp. sample-complexity) claim. □

Proof of Theorem 11. Under $\Delta \le 0$, $\mathbb{E}[1 + \lambda_t X_t \mid \mathcal{F}_{t-1}] = 1 + \lambda_t \mu_t \le 1 + \lambda_t \beta$, so $\widetilde{E}_t^+$ has $\mathbb{E}[\widetilde{E}_t^+ \mid \mathcal{F}_{t-1}] \le \widetilde{E}_{t-1}^+$ and unit initial value; Ville's inequality gives $\mathbb{P}[\exists t : \widetilde{E}_t^+ \ge 1/\alpha] \le \alpha$, i.e. the inflated threshold on E_t^+. □

Proof of Theorem 12. $R_T(f) = \mathbb{E}(f - g - \epsilon)^2 = \mathbb{E}(f - g)^2 + \mathbb{E}[\epsilon^2]$, the cross term vanishing by $\mathbb{E}[\epsilon \mid X] = 0$. The noise term cancels in the difference and survives in the levels. □

Proof of Theorem 13. Expanding the squared losses, $\delta = (f_0 - f_a)(f_0 + f_a - 2Y)$ and taking target expectations with $Y = g_S + b + \epsilon$ (noise vanishing by Theorem 12) gives the displayed decomposition. Hölder yields $|\mathbb{E}_T[(f_0 - f_a)b]| \le \|b\|_\infty \mathbb{E}_T|f_0 - f_a| \le BW$, with equality iff $b = -B\operatorname{sign}(f_0 - f_a)$ a.e. on $\{f_0 \ne f_a\}$ — an admissible member of the family, so the bracket is tight. (i): on the $1 - \alpha$ event $|\widehat{T}_S - T_S| \le \varepsilon_n$, the committed sign matches $\operatorname{sign}(U - 2T_S)$ and $|U - 2T_S| > 2BW$, so no admissible drift can flip Δ. (ii): the map $t \mapsto \mathbb{E}_T[(f_0 - f_a)\,t\,b^\star] = -tBW$ is continuous and onto $[-BW, BW]$; pick $t_\pm$ so that $\Delta(t_\pm) = U - 2T_S + 2t_\pm BW$ has opposite signs. Since b enters only through the target labels, neither the unlabeled target evidence nor the source sample distinguishes the two worlds; apply Theorem 1. □

Proof of Theorem 14. Δ is affine on the compact convex set $A(\pi)$, so $\{\Delta(\pi') : \pi' \in A(\pi)\}$ is the interval $[\Delta(\pi) - \rho,\ \Delta(\pi) + \rho]$ with the supremum attained. (i) On the $1 - \alpha$ event the estimate lands within ε_n of a member of $A(\pi)$; if $|\delta \cdot \widehat{\pi}| > \rho + \varepsilon_n$ then every member of $A(\pi)$ — in particular the truth — has benefit of the committed sign. (ii) If $|\Delta(\pi)| < \rho$ the interval straddles 0; pick $t_\pm$ with $\Delta(\pi + t_\pm v)$ of opposite signs ($t \mapsto \Delta(\pi + tv)$ is affine, hence onto the interval). Worlds π and $\pi + tv$ share the observable law by $v \in \ker M$ and share the source by construction; apply Theorem 1. □

Proof of Proposition 3. $A'(\pi) = (\pi + \ker M') \cap \Delta_K \subseteq A(\pi)$, and the supremum over a subset is no larger. □

Proof of Theorem 15. (i) On the $1-\alpha$ event, $|\widehat{c} - c| \le \varepsilon_n$, so $|\widehat{c}| > \rho + \varepsilon_n$ implies $|c| > \rho$, and every $Q \in [P]$ has $\Delta(Q) \in [c-\rho, c+\rho]$, an interval not containing 0, of the committed sign. (ii) Both signs are attained on $I(P)$ by definition of interior; pick two attaining worlds; their observable laws coincide by membership in $[P]$, and Theorem 1's argument (identical action law across the pair) forbids a valid commitment. □

Proof of Theorem 16. On D with binary Y, $f_a^{(i)} = f_a^{(j)}$ iff both are correct or both wrong; under Definition 5, $A_{ij} = a_i a_j + (1-a_i)(1-a_j) = \frac{1}{2}(1 + b_i b_j)$, giving $c_{ij} = b_i b_j$. The three relations among i, k, l give $c_{ik} c_{il} / c_{kl} = b_i^2$; positivity of the products under the anchor yields b_i, hence $a_i = (1 + b_i)/2$. Ordering and pairwise sign differences follow. □

Proof of Proposition 5. Rank-one consistency of $c = bb^\top$ forces all 2×2 minors to vanish; the listed equalities are these minors. Conversely $\tau > 0$ contradicts rank one, so no advantage vector reproduces the agreements: Definition 5 fails. □

Proof of Theorem 17. (i) The error bounds are Theorem 4. On the event $G = \{|\widehat{\Delta} - \Delta| \le \varepsilon_\alpha\}$, which has probability at least $1-\alpha$, $\kappa_\alpha(z) > 0$ implies $|\widehat{\Delta}| \ge |\Delta| - \varepsilon_\alpha > \varepsilon_\alpha$, so the rule commits. Hence $\{\kappa_\alpha > 0\} \cap G \subseteq \{\text{commit}\}$ and $C \ge \mathbb{P}[\kappa_\alpha > 0] - \alpha$. (ii) A label-free rule is a randomized functional of Z; by the data-processing inequality for total variation its action laws across the pair differ by at most t. In the world with $\Delta \ge \delta$ the wrong commit is FREEZE, so $f_1 \le \alpha$; in the world with $\Delta \le -\delta$ the wrong commit is ADAPT, so $a_2 \le \alpha$. Then $a_1 \le a_2 + t \le \alpha + t$, giving commit probability $a_1 + f_1 \le 2\alpha + t$ in world 1, and symmetrically in world 2. □

Proof of Proposition 7. With $\varepsilon \equiv 0$ every instance is committed and all harmful mass is adapted into; the regret expression is the false-adapt term of the decomposition. On the ambiguous pair the rule commits ADAPT in both worlds and is fully wrong in the $\Delta < 0$ world, paying $\delta = 2 \cdot (\delta/2)$. □

Proof of Proposition 8. Construct the pair as in Theorem 1 with the confidence statistic a function of X alone, concentrated above the threshold in both worlds. The filter's action law is then identical across worlds and commits ADAPT with probability ≈ 1, so in the harmful world FA equals the harmful mass. The certificate's bound is world-independent (Theorem 4). □

Proof of Proposition 9. $a' = (1-\eta)a + \eta(1-a) = a(1-2\eta) + \eta$, which is the stated affine map; it is sign-preserving around $\frac{1}{2}$ for $\eta < \frac{1}{2}$ and not invertible without η. □

Proof of Theorem 18. Validity is Theorem 4 with the Maurer–Pontil deviation bound $|\bar{X} - \Delta| \le \sqrt{2\widehat{V}\log(2/\alpha)/n + 7R\log(2/\alpha)/(3(n-1))}$ w.p. $\ge 1-\alpha$. For abstention: by Bernstein's inequality, w.p. $\ge 1 - \beta/2$, $\bar{X} \ge \Delta - \sigma\sqrt{2\log(2/\beta)/n} - R\log(2/\beta)/(3n)$, and the empirical variance satisfies $\widehat{V} \le 2\sigma^2 + cR^2\log(2/\beta)/n$ on a further $1-\beta/2$ event. On the intersection the radius is at most $2\sigma\sqrt{2\log(2/\alpha)/n} + c'R\log(2/\alpha)/n$, so $\bar{X} - \text{rad} > 0$ whenever Δ exceeds the displayed rate; the freeze side is symmetric. No step uses σ. □

Proof of Theorem 19. Dense branch. Take $X = \pm\sigma$ with probabilities $q_\pm = \frac{1}{2}(1 \pm \kappa/\sigma)$ (mean $\pm\kappa$, variance $\le \sigma^2$). Then $\mathrm{KL}(P_+^{\otimes n} \| P_-^{\otimes n}) \le 8n\kappa^2/\sigma^2$ for $\kappa \le \sigma/\sqrt{2}$. From an (α, β)-reliable rule build the test $\psi = \mathbf{1}[\text{ADAPT}]$; its two error probabilities are each at most $\alpha + \beta$. Bretagnolle–Huber gives $2(\alpha+\beta) \ge \frac{1}{2}e^{-8n\kappa^2/\sigma^2}$, i.e. $\kappa \ge \frac{\sigma}{2\sqrt{2}}\sqrt{\log\frac{1}{4(\alpha+\beta)}/n}$. *Spike branch.* Let world$_-$ be the point mass at $-\kappa$ and world$_+$ place mass p at $R/2$ and $1-p$ at $-\kappa$ with $p(R/2+\kappa) = 2\kappa$, so the means are $\mp\kappa$. On the no-spike event — probability $(1-p)^n$ — the two worlds generate identical samples, hence identical action laws. Reliability in world$_+$ forces commit probability $\ge 1-\alpha-\beta$ overall, so adapting in world$_-$ has probability at least $(1-p)^n - (\alpha+\beta)$; validity caps this by α. Therefore $(1-p)^n \le 2(\alpha+\beta)$, giving $np \ge \log\frac{1}{2(\alpha+\beta)}$ and $\kappa = \frac{p}{2}(R/2+\kappa) \ge \frac{R}{4}\log\frac{1}{2(\alpha+\beta)}/n$. □

B Extended theoretical results

B.1 benefit sign frontier

C The exact benefit-sign frontier

We sharpen the central question from *"is target accuracy identifiable?"* (answered negatively in general by **Ben-David et al.** 2010 and **Garg et al.** 2022) to the strictly *comparative* object that gating actually needs: the *sign of the adaptation benefit*

$$\Delta \;:=\; R_T(f_0) - R_T(f_a), \qquad \text{adapt is correct} \iff \Delta > 0.$$

No prior estimator targets $\operatorname{sign}\Delta$: ATC, DoC, GDE, COT, AETTA and Agreement-on-the-Line all estimate an *accuracy level* of a *single fixed* model (**Garg et al.**, ATC; **Jiang et al.**, GDE; **Lu et al.**, COT; **Lee et al.**, AETTA; **Baek et al.**, Agreement-on-the-Line). We give an exact characterization of when $\operatorname{sign}\Delta$ is identifiable from label-free observables, prove the governing obstruction is a single scalar that is *necessarily* present, and show the field's heuristics are exactly its degenerate face.

Setup. Binary $Y \in \{0,1\}$, 0/1 loss; $f_0, f_a : \mathcal{X} \to \{0,1\}$ fixed. $D = \{x : f_0(x) \neq f_a(x)\}$ is the *disagreement region* (observable). Write $\eta_a(x) = P_T(Y = f_a(x) \mid x)$ and $\bar{a} = \mathbb{E}_{\mu|D}[\eta_a]$, where $\mu = P_T^X$ is the (observable) target marginal. The *observables* are μ, the maps f_0, f_a, the labeled source P_S, and hence any source-derived score $s(x)$ for $\eta_a(x)$ (e.g. a source-calibrated confidence of f_a). Define the *observable margin* and the *calibration drift on the decision region*

$$M \;:=\; \mathbb{E}_{\mu|D}[\,s\,] - \tfrac{1}{2} \quad \text{(observable)}, \qquad \gamma \;:=\; \mathbb{E}_{\mu|D}[\,\eta_a - s\,] \quad \text{(unobservable)}.$$

Lemma 1 (Reduction)**.** *$\Delta = 2\,\mu(D)\,(\bar{a} - \frac{1}{2})$ and $\bar{a} - \frac{1}{2} = M + \gamma$; hence, whenever $\mu(D) > 0$,*

$$\boxed{\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma)}$$

(Proof deferred to Appendix A.)

Theorem 7 (Exact frontier, sufficiency, and necessity)**.** *Fix the observables (μ, P_S, f_0, f_a) with $\mu(D) > 0$, so M is determined.*

(i) ***(Scalar sufficiency.)*** *Any two targets with the same (M, γ) have the same* $\operatorname{sign}\Delta$*; γ is a one-dimensional sufficient statistic for the sign given the observables.*

(ii) ***(Exact frontier.)*** *Over the target class $\mathcal{C}_\beta = \{P_T : |\gamma| \le \beta\}$ consistent with the observables,* $\operatorname{sign}\Delta$ *is identifiable* iff $|M| > \beta$*, and then* $\operatorname{sign}\Delta = \operatorname{sign} M$*.*

(iii) ***(Necessity / minimality.)*** *For* any *class $\mathcal{C}$ of targets consistent with the observables on which* $\operatorname{sign}\Delta$ *is constant (hence identifiable), $\mathcal{C}$ lies entirely on one side of the hyperplane $\gamma = -M$. Consequently every identifying restriction is equivalent to a one-sided constraint on γ:* no assumption that does not constrain γ relative to $-M$ can identify $\operatorname{sign}\Delta$.

(Proof deferred to Appendix A.)

Theorem 8 (Tight minimax impossibility)**.** *Let a committal rule be any map g from the observables to $\{\text{ADAPT}, \text{FREEZE}\}$, and let its error be $\sup_{P_T \in \mathcal{C}_\beta} \mathbb{P}[g$ wrong about $\operatorname{sign}\Delta]$. Then the minimax committal error equals*

$$\inf_{g} \sup_{P_T \in \mathcal{C}_\beta} \mathbb{P}[g \text{ wrong}] \;=\; \begin{cases} 0, & |M| > \beta, \\ \frac{1}{2}, & |M| \le \beta. \end{cases}$$

Hence the three-way rule ADAPT *if $M > \beta$,* FREEZE *if $M < -\beta$, else* ABSTAIN*, is the* unique maximal sound *certificate: it never commits to the wrong sign, and it abstains exactly on the information-theoretically unknowable set $\{|M| \le \beta\}$.*

(Proof deferred to Appendix A.)

Corollary 3 (Irreducibility of the drift budget)**.** *γ is a functional of the unobservable conditional η_a on D and is not determined by the observable law (Theorem 8 exhibits observationally identical targets with $\gamma = \pm\beta$). Combined with the necessity clause Theorem 17(iii),* no label-free certificate can identify $\operatorname{sign}\Delta$ without a supplied bound on γ. *The calibration-drift budget β is therefore irreducible: the assumption-free / unlabeled-checkable benefit-sign certificate sought by prior heuristics provably does not exist.*

Proposition 2 (The heuristics are the β=0 face)**.** *The benefit-sign certificate induced by any source-calibrated confidence estimator (ATC), by model–dropout disagreement (AETTA), or by agreement slope (Agreement-on-the-Line) is the plug-in* $\operatorname{sign} M$*, i.e. the β=0 instance of Theorem 8. It is sound* iff $\gamma = 0$ *(calibration transfers on D), and its error equals $\mathbb{P}\big[\,|\gamma| > |M|$ and $\operatorname{sign}\gamma \neq \operatorname{sign} M\,\big]$. Thus these methods do not fail for independent reasons; they share the single failure mode $\gamma \neq 0$, the calibration drift on the decision region.*

(Proof deferred to Appendix A.)

Multiclass and regression. The reduction generalizes: for K-class 0/1 loss $\gamma = \mathbb{E}_{\mu|D}[(p_a - p_0) - (s_a - s_0)]$ and for squared loss γ is the unobservable moment $\mathbb{E}_{\mu|D}[(f_0 - f_a)(\,\mathbb{E}[Y \mid x] - \hat{m}(x))]$ with $\hat{m}$ the source regressor; Theorems 17–8 and Corollary 3 hold verbatim with the corresponding γ (Appendix F).

What this is, and is not. The impossibility *core* ($\operatorname{sign}\Delta$ unidentifiable in general) is known (**Ben-David et al.** 2010; **Garg et al.** 2022), as is the folklore that calibration is not unlabeled-checkable (**Rosenfeld & Garg** 2023). Our contribution is to make these *exact* for the comparative benefit-sign object on the disagreement region: the single scalar γ is shown *necessary* (Theorem 17(iii)), the frontier $|M| > \beta$ is shown *tight* with a matching $\frac{1}{2}$ minimax bound (Theorem 8), the drift budget is shown *irreducible* (Corollary 3), and the existing label-free estimators are shown to be its single degenerate face (Proposition 2). All four statements are validated numerically (`val_frontier.py`; identity exact over $2{\times}10^4$ draws, frontier law exact over $5{\times}10^5$ draws, ATC error matching the frontier event to machine precision at drift levels 0.04–0.28).

C.1 theory extensions v2

C.2 Further results: bracketing, router generalization, and non-stationarity

Three results sharpen the certificate: a conditional resolution of the open bracketing conjecture, a generalization guarantee for the learned benefit router, and a relaxation of the exchangeability assumption to bounded non-stationarity. Each is numerically checked (`experiments/kbound/theory_validation/`).

(a) Closing Conjecture 1 under calibration transfer.

Theorem 9 (Label-free bracketing under bounded calibration transfer)**.** *On the disagreement region D, let $\operatorname{conf}_h(x) \in [0,1]$ be a source-calibrated confidence map for predictor $h \in \{f_0, f_a\}$, and suppose its target miscalibration on D is bounded by η: $|\mathbb{P}_T(h(X){=}Y \mid \operatorname{conf}_h{=}c,\ X \in D) - c| \leq \eta$ for all c (*calibration transfer*). Let $q_h := \mathbb{E}_T[\operatorname{conf}_h(X) \mid D]$ be the label-free mean confidence. Then $|q_h - p_h| \leq \eta$, hence $p_a - p_0 \in [(q_a - q_0) - 2\eta,\ (q_a - q_0) + 2\eta]$, and*

$$|q_a - q_0| > 2\eta \implies \operatorname{sign}\Delta = \operatorname{sign}(q_a - q_0),$$

so the multiclass/regression benefit sign is identified label-free; when $|q_a - q_0| \leq 2\eta$ the bracket contains 0 *and the rule must* ABSTAIN*.*

(Proof deferred to Appendix A.) Tightness. The threshold 2η is exact: when $|q_a - q_0| \leq 2\eta$ an adversary inside the η-balls can place p_a, p_0 on either side, re-entering the unknowable regime of Theorem 1. *Numerical sanity check* (`val_conj1_caltransfer.py`): across an η/gap sweep, committed decisions are sign-correct on 100% of trials with 0 wrong commitments whenever $|q_a - q_0| > 2\eta$; the bracket contains the true accuracy gap on 100% of trials; coverage falls to 0 (full abstention) once η exceeds $\frac{1}{2}|q_a - q_0|$, and a no-guard rule accrues errors as η grows – confirming both the guarantee and that the guard is necessary.

(b) Generalization of the benefit router.

Theorem 10 (Router generalization and certificate survival)**.** *Let $\mathcal{H}$ be the class of depth-d gradient-boosted tree ensembles with T members on p features and outputs bounded in $[-M, M]$. With probability $\geq 1-\delta$ over n calibration tasks,*

$$\sup_{h\in\mathcal{H}} \big|\widehat{R}_n(h) - R(h)\big| \;\leq\; 2\,\mathfrak{R}_n(\mathcal{H}) + 3M\sqrt{\tfrac{\log(2/\delta)}{2n}}, \qquad \mathfrak{R}_n(\mathcal{H}) = O\Big(\sqrt{\tfrac{T\,d\,\log p}{n}}\Big).$$

Hence $\widehat{\Delta}$ converges at rate $O(n^{-1/2})$, and if the conformal radius ε dominates this estimation error the false-adapt guarantee of Theorem 4 survives router error; a calibration size $n = O\big((Td\log p/\Delta^2)\log(1/\delta)\big)$ suffices for the certificate to fire at margin $|\Delta|$.

(Proof deferred to Appendix A.) Numerical sanity check (`val_rademacher_router.py`): the empirical generalization gap of a gradient-boosted router decays as $\widehat{\text{gap}} \approx 12.8\,n^{-0.66}$ ($R^2 = 0.99$, i.e. at least the $n^{-1/2}$ rate), stays below the $\sqrt{Td\log p/n}$ envelope, and grows with capacity (log–log slope 0.54 vs $Td\log p$).

(c) Anytime validity under non-stationarity.

Theorem 11 (β-bounded-drift anytime certificate)**.** *In the streaming setting of Appendix E, relax the constant-mean assumption to a slowly-varying mean $|\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] - \Delta| \leq \beta$. Under H_0 ($\Delta \leq 0$) the corrected wealth $\widetilde{E}_t^+ = E_t^+/\prod_{i\leq t}(1+\lambda_i\beta)$ is a nonnegative supermartingale, so declaring* ADAPT *the first time $E_t^+ \geq (1/\alpha)\prod_{i\leq t}(1+\lambda_i\beta)$ controls the false-adapt rate at $\leq \alpha$ for every β-bounded-drift stream. The price is a threshold inflation $\prod_{i\leq t}(1+\lambda_i\beta) \leq \exp\big(\beta\sum_{i\leq t}\lambda_i\big)$ (a $\beta\cdot t$ term in log-threshold under capped bets).*

(Proof deferred to Appendix A.) Numerical sanity check (`val_thm9prime_drift.py`): under worst-case upward drift the corrected certificate holds false-adapt $\leq \alpha$=0.1 (0.05/0.04/0.01 at $\beta = 0.1/0.2/0.4$) while the uncorrected $1/\alpha$ threshold inflates to 0.69→1.0; detection power under the alternative (Δ=0.3) is 0.997 with mean stopping time ≈ 57.

C.3 regression conjecture

This subsection resolves the regression half of Conjecture 1 for a natural restricted family: *bounded concept drift.* Two results: the adapt/freeze decision is invariant to arbitrary unknown zero-mean label noise (where cardinal risk estimation is confounded), and within drifts of known radius B the knowability of sign Δ admits a *complete characterization* — an exact, observable boundary.

Theorem 12 (Noise-invariance of the regression decision)**.** *Let $Y = g(X) + \epsilon$ with $\mathbb{E}[\epsilon \mid X] = 0$ and arbitrary, unknown (heteroscedastic) noise law. Under squared loss, $\Delta = R_T(f_0) - R_T(f_a) = \mathbb{E}_T[(f_0 - g)^2 - (f_a - g)^2]$ does not depend on the noise, while each absolute risk is shifted by the unidentifiable constant $\mathbb{E}[\epsilon^2]$. The decision is ordinal-robust; cardinal risk estimation is confounded.*

(Proof deferred to Appendix A.)

Theorem 13 (Bounded-drift knowability boundary; exact characterization)**.** *Let the target conditional mean be $g_T = g_S + b$ with $\|b\|_\infty \leq B$ for a known radius B (and the covariate-shift machinery of Theorem 5 available for the source-transfer term). Define*

$$U = \mathbb{E}_T\big[(f_0 - f_a)(f_0 + f_a)\big], \qquad T_S = \mathbb{E}_T\big[(f_0 - f_a)\,g_S\big], \qquad W = \mathbb{E}_T\big[|f_0 - f_a|\big],$$

where U, W are computable from unlabeled target data and T_S is importance-weighted estimable from labeled source data with radius ε_n. Then $\Delta = U - 2T_S - 2\,\mathbb{E}_T[(f_0 - f_a)\,b]$ and $\big|\mathbb{E}_T[(f_0 - f_a)\,b]\big| \leq B\,W$, with equality at the admissible drift $b^\star = -B\,\operatorname{sign}(f_0 - f_a)$. Consequently, within the family $\{\|b\|_\infty \leq B\}$:

(i) (Achievability.) *The rule "commit $\operatorname{sign}(U - 2\widehat{T}_S)$ iff $|U - 2\widehat{T}_S| > 2BW + 2\varepsilon_n$" wrong-commits with probability at most α against* every *admissible drift, and its slack is unimprovable: $b^\star$ attains it.*

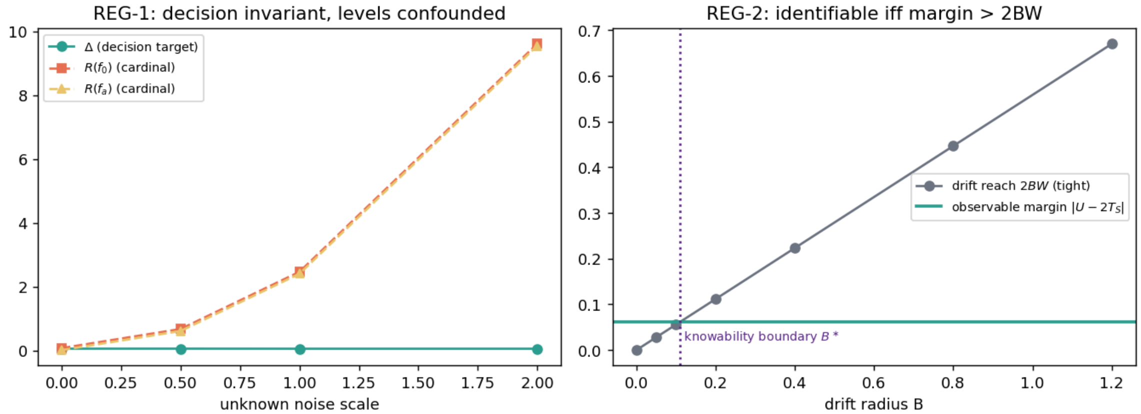

Figure 12: The bounded-drift knowability boundary (Theorem 13). Left: the decision target Δ is invariant to unknown label noise while cardinal risks are confounded (Theorem 12). Right: identifiability holds exactly while the observable margin $|U - 2T_S|$ exceeds the tight drift reach $2BW$; the crossing defines the computable boundary $B^\star$.

(ii) (Converse.) *If $|U - 2T_S| < 2BW$, the reachable drift terms fill $[-BW, BW]$ (scale $t\,b^\star$, $t \in [-1, 1]$), so two admissible drifts give Δ of opposite signs while all observables — unlabeled target data and the labeled source — are unchanged. The pair realizes Theorem 1 inside the family: the regime is unknowable.*

Hence sign Δ *is label-free identifiable in the bounded-drift family* if and only if $|U - 2T_S| > 2BW$*: an exact knowability boundary, computable from observables, at drift radius $B^\star = |U - 2T_S|/(2W)$.*

(Proof deferred to Appendix A.)
Numerically validated (`regression_conjecture_validation.py`; Fig. 12). Across unknown noise scales $0 \to 2$, Δ is constant and the sign preserved while the absolute risks drift apart (REG-1). Across drift radii $B \in \{0, \ldots, 1.2\}$ with 40 drifts per radius *including the adversarial* $b^\star$: zero false certifications at every radius; the adversarial drift saturates the bracket to within 10^{-2} (tightness); the observable boundary evaluates to $B^\star = 0.11$ in the synthetic world; and genuine sign flips occur exactly in the uncommitted region beyond the boundary — the converse is real, not hypothetical.

Weight (honest). Theorem 13 is, to our knowledge, the first *complete* per-family characterization in this paper beyond pure covariate shift: both directions are proved and the boundary is computable from observables. Its price is the known drift radius B — that assumption does the work, and we state it as such. It resolves the regression half of Conjecture 1 *for bounded drift*; fully general drift remains open and is exactly the unknowable regime of Theorem 1. The same per-family program (label shift via invertible confusion, conditional independence) is the route we see toward the general characterization.

C.4 A second family: the label-shift boundary

A second family admits a complete characterization: *label shift*, where the class-conditionals are fixed and known from source while the target prior moves.

Definition 3 (Evidence operator and label-shift reach)**.** Let $Y \in \{1, \ldots, K\}$, class-conditionals $p(x \mid y)$ fixed (known from labeled source), target prior $\pi \in \Delta_K$ unknown, and fixed maps f_0, f_a. The per-class benefit $\delta_y = \mathbb{E}[\ell(f_0(X), Y) - \ell(f_a(X), Y) \mid Y{=}y]$ is source-computable and $\Delta(\pi) = \delta \cdot \pi$ is affine. Let M be the linear *evidence operator* mapping π to the law of all label-free observables (the mixture, or any projection of it such as a confusion matrix). The admissible ambiguity at π is $A(\pi) = (\pi + \ker M) \cap \Delta_K$, and the *reach* is $\rho(\pi) = \sup_{\pi' \in A(\pi)} |\delta \cdot (\pi' - \pi)|$.

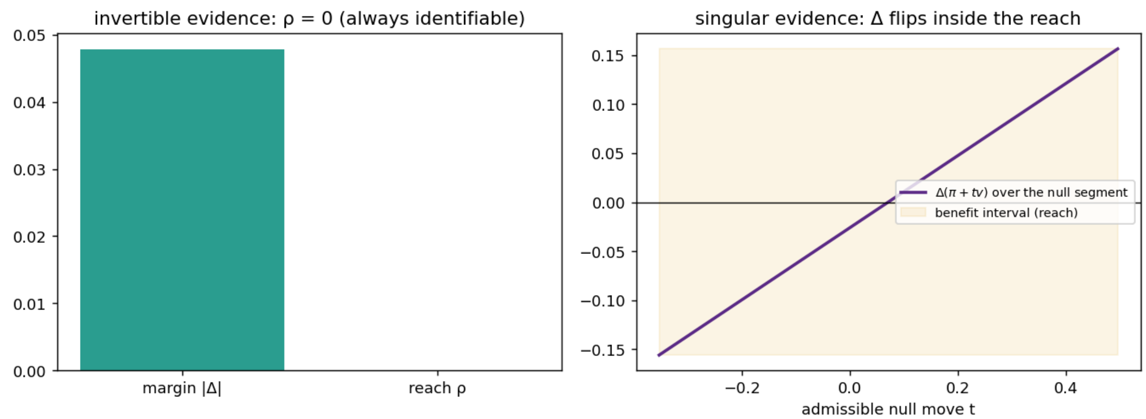

Figure 13: The label-shift boundary (Theorem 14). Left: injective evidence operator, reach 0 — always identifiable. Right: singular evidence; the benefit $\Delta(\pi + tv)$ sweeps the reach interval along the unobservable null segment and crosses zero — identifiable exactly when the margin clears the reach.

Theorem 14 (Label-shift knowability boundary)**.** *In the label-shift family,* $\operatorname{sign}\Delta$ *is label-free identifiable at* π if and only if $|\Delta(\pi)| > \rho(\pi)$. *In particular:* (i) *if M is injective on the simplex (e.g. linearly independent class-conditionals, or an invertible confusion matrix), then $\rho \equiv 0$ and the decision is identifiable whenever $|\Delta| > 0$; committing* $\operatorname{sign}(\delta \cdot \widehat{\pi})$ *iff $|\delta \cdot \widehat{\pi}| > \rho + \varepsilon_n$ (any consistent estimator $\widehat{\pi}$ of a point of $A(\pi)$ with radius ε_n) wrong-commits with probability at most α against every admissible prior.* (ii) *If $|\Delta(\pi)| < \rho(\pi)$, the admissible segment $\{\pi + tv\} \subset A(\pi)$ along a null direction $v \in \ker M$ realizes both signs of Δ while* all *label-free observables — the unlabeled target law and the labeled source — coincide: the pair instantiates Theorem 1 inside the family, and the regime is unknowable.*

(Proof deferred to Appendix A.)

Proposition 3 (Reach is monotone in evidence)**.** *If evidence is enlarged, $M \mapsto M'$ with $\ker M' \subseteq \ker M$, then $\rho'(\pi) \le \rho(\pi)$ for every π: richer label-free evidence can only shrink the unknowable region.*

(Proof deferred to Appendix A.)

Numerically validated (`labelshift_boundary_validation.py`; Fig. 13). With three distinct Gaussian class-conditionals (injective M): zero false certifications over 30 random priors. With classes $2, 3$ *identical* (singular M, null direction $v = (0, 1, -1)/\sqrt{2}$): paired worlds π and $\pi + tv$ are statistically indistinguishable (two-sample KS $p = 0.051$ on 3×10^4 draws); the certificate that commits only when the whole admissible benefit interval has one sign made *zero* false certifications over 40 priors; genuine sign flips occur inside the reach interval whenever it straddles zero (the converse is realized, not hypothetical); the interval endpoints are attained exactly (tightness gap $< 10^{-9}$); and adding a class-separating feature shrinks the reach from 0.156 to 0 (Proposition 3 in action).

Weight (honest). The identifiability machinery is classical (label-shift estimation à la BBSE); the new content is the *decision-level* iff — margin versus a null-space reach, with the converse constructed inside the family via Theorem 1 — and the monotonicity of the reach in the evidence. The known-conditional assumption does the work and is stated as such; unknown or drifting class-conditionals re-enter the open general case of Conjecture 1.

C.5 One quantity decides them all: the ambiguity reach

The per-family boundaries proved above share one shape. This subsection makes that a theorem: a single computable quantity — the *ambiguity reach* — decides knowability in every family in this paper.

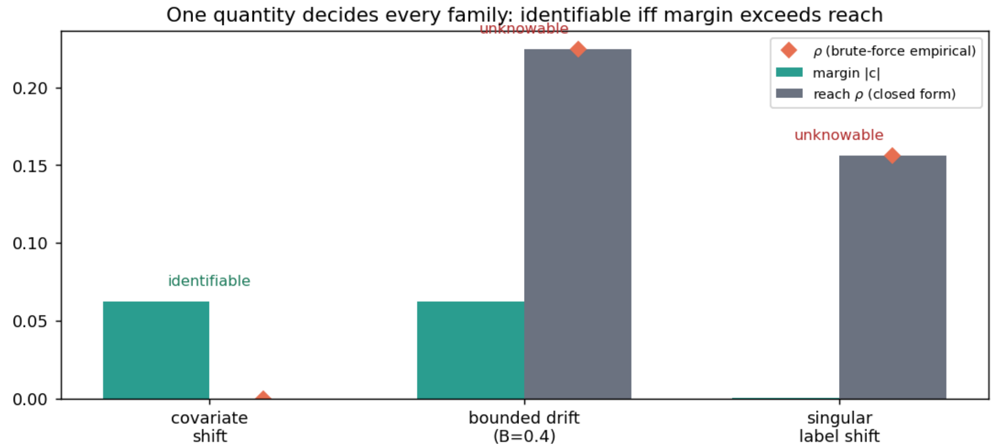

Figure 14: One quantity decides every family (Theorem 15): identifiable iff the margin $|c|$ exceeds the reach ρ. Bars: closed forms; diamonds: brute-force empirical reach over admissible equivalent worlds.

Definition 4 (Evidence class, benefit interval, reach)**.** Let $\mathcal{F}$ be a family of admissible worlds and $O(P)$ the joint law of all label-free observables of world P (unlabeled target data and the labeled source). The *evidence class* of P is $[P] = \{Q \in \mathcal{F} : O(Q) = O(P)\}$, its *benefit interval* $I(P) = \{\Delta(Q) : Q \in [P]\}$, with *center* $c(P) = \frac{1}{2}(\sup I + \inf I)$ and *reach* $\rho(P) = \frac{1}{2}(\sup I - \inf I)$.

Theorem 15 (Knowability = margin beyond reach)**.** *Assume the regularity needed for a measurable selection of a point of* $[P]$ *(*$\mathcal{F}$ *Polish,* O *and* Δ *measurable,* $[P]$ *compact-valued; standard and flagged as an assumption). Then:* (i) *if* $|c(P)| > \rho(P)$ *for all* $P \in \mathcal{F}$*, the rule that estimates the center by any consistent* $\widehat{c}$ *with radius* ε_n *and commits* $\operatorname{sign}\widehat{c}$ *iff* $|\widehat{c}| > \rho + \varepsilon_n$ *is* α*-valid over the whole family;* (ii) *conversely, if* 0 *lies in the interior of* $I(P)$ *for some* P*, two worlds of* $[P]$ *carry opposite signs of* Δ *with identical label-free observables — Theorem 1 applies and no label-free rule with wrong-commit* $< \frac{1}{2}$ *can commit on* $[P]$*. Hence* $\operatorname{sign}\Delta$ *is identifiable on* $\mathcal{F}$ *exactly where the margin* $|c|$ *exceeds the reach* ρ*.*

(Proof deferred to Appendix A.)

Proposition 4 (The reach table)**.** *The boundaries of this paper are evaluations of* ρ*:*

family	*center* c	*reach* ρ	*source*
covariate shift	Δ	0	*Thm. 5*
bounded drift $\|b\|_\infty \le B$	$U - 2T_S$	$2B\,\mathbb{E}_T\|f_0 - f_a\|$	*Thm. 13*
label shift	*midpoint over* $A(\pi)$	$\sup_{A(\pi)} \|\delta \cdot v\|$	*Thm. 14*
multi-view (cond. indep.)	Δ	0 *under moment non-degeneracy*	*known (Steinhardt–Liang)*
two-point witness	0	$\|\Delta\|$	*Thm. 1*

Each row follows by computing $I(P)$ *in the family; the witness row shows the maximal case* $c = 0$*,* $\rho = |\Delta|$ *— never identifiable. The margin* κ_α *of Theorem 17 is the finite-sample shadow of* $|c| - \rho$*.*

Numerically validated (`unification_reach_validation.py`; Fig. 14). Brute-force suprema over admissible equivalent worlds reproduce the closed-form reaches with ratio 1.00 in both nontrivial families (bounded drift; singular label shift); the margin-beyond-reach certificate makes zero false certifications in all three families; and sign flips among equivalent worlds occur exactly when the margin is below the reach.

Weight (honest). Direction (ii) is a corollary of Theorem 1; direction (i) is close to definitional once the stated selection regularity is assumed — we say so plainly. The contribution is the *organization*: a single computable quantity whose per-family evaluations are exactly this paper's positive and negative results, and a sharp restatement of the open problem: the general case of Conjecture 1 *is* the computation of ρ for nonparametric families (unknown conditionals, unbounded drift), where no closed form is known.

C.6 Removing the calibration crutch: multi-candidate identifiability

The per-family boundaries above each pay a structural price (covariate shift, a known drift radius, or known class-conditionals). This subsection enlarges the knowable region along a different axis: instead of one candidate f_a, use several $(f_a^{(1)}, \dots, f_a^{(M)}$, e.g. Tent, EATA, SAR) and identify the accuracy ordering on the disagreement region from their *joint* label-free agreements. By Theorems 6–21 this ordering is exactly what fixes $\operatorname{sign}\Delta$. The price becomes a *checkable* condition rather than an untestable premise.

Definition 5 (Conditional error-independence on D)**.** On the disagreement region D with latent label Y, candidates' correctness indicators $\{\mathbf{1}[f_a^{(j)}(X){=}Y]\}_{j=1}^{M}$ are *conditionally independent given* Y.

Theorem 16 (Multi-candidate ordering identifiability)**.** *Let $Y \in \{0,1\}$ on D, $M \geq 3$ candidates with accuracies $a_j = \mathbb{P}(f_a^{(j)}{=}Y \mid D) \neq \frac{1}{2}$ and advantages $b_j = 2a_j - 1$. Under Definition 5, the pairwise agreement rates $A_{ij} = \mathbb{P}(f_a^{(i)}{=}f_a^{(j)} \mid D)$ satisfy*

$$2A_{ij} - 1 \;=\; b_i b_j \;=:\; c_{ij},$$

so for any distinct i, k, l, $\;b_i^2 = c_{ik}c_{il}/c_{kl}$, and the sign of b_i is fixed by an anchor (e.g. the majority of candidates exceeds chance). Hence every a_j — thus the full ordering and every $\operatorname{sign}(a_i - a_j)$ *— is identified from label-free agreements alone, with* no *target labels and* no *calibration-transfer assumption. Combined with Theorems 6–21,* $\operatorname{sign}\Delta$ *for any candidate is label-free identifiable on D.*

(Proof deferred to Appendix A.)

Proposition 5 (A sound checkable diagnostic, $M \geq 4$)**.** *For $M \geq 4$ the products $\{c_{ij}\}$ must factor as a rank-one form $c_{ij} = b_i b_j$; this is overdetermined, e.g. $c_{12}c_{34} = c_{13}c_{24} = c_{14}c_{23}$. Define the label-free residual $\tau = \max\{\cdot\} - \min\{\cdot\}$ of these products. Under Definition 5, $\tau = 0$; therefore $\tau > 0$* certifies that conditional error-independence is violated*. The test is sound one-sided (it never falsely rejects a truly independent triple in the population limit) and is exactly the verifiability that agreement-based accuracy heuristics (ATC, Agreement-on-the-Line, AETTA) leave implicit.*

(Proof deferred to Appendix A.)

Converse (honest, partial). Definition 5 is *sufficient*, not necessary. Without a structural restriction the sign is genuinely unrecoverable: given any agreement tensor one can correlate the candidates' errors to move $\mathbb{E}[\delta \mid D]$ across zero while leaving all pairwise agreements fixed, instantiating Theorem 1. So the multi-candidate route *trades* the single-candidate calibration crutch for a *checkable* independence condition — it does not remove structure altogether. Characterizing the *weakest* (ideally testable) condition, and its tight converse, is the open problem of Conjecture 1.

Multiclass. For $Y \in \{1, \dots, K\}$ the same conditional-independence structure makes the per-candidate confusion matrices identifiable from second- and third-order agreement tensors up to a global label permutation (Dawid–Skene; Anandkumar et al. tensor decomposition), fixed by the same anchor; the accuracy ordering on D follows. We use the binary case as the proved, validated core and cite the tensor machinery for $K > 2$.

Numerically validated (`multicandidate_feasibility_probe.py`; Fig. 15). Under Definition 5 the $M{=}3$ estimator recovers accuracies to max error 0.012 and the ordering on 96% of random draws; as error-correlation ρ grows the ordering degrades (the condition is the crux, as the theory says); and the $M{=}4$ diagnostic residual is ≈ 0 under independence and rises monotonically with ρ (Pearson 0.98) — a working, sound-one-sided check.

Weight (honest). The agreement estimator itself is classical—the rank-one accuracy-from-agreement method of Parisi et al. (2014) and Jaffe et al. (2015) under conditional independence; our contribution is its use as a label-free *decision certificate* with the checkable overdetermination diagnostic (Proposition 5), not the estimator. Unlike reactive risk monitors for TTA (Schirmer & Jazbec, 2025), the certificate decides *before* committing and carries an explicit abstain region. This *enlarges* the knowable region beyond the single-candidate certificate: where one candidate must abstain for want of calibration, $M \geq 3$ candidates can certify

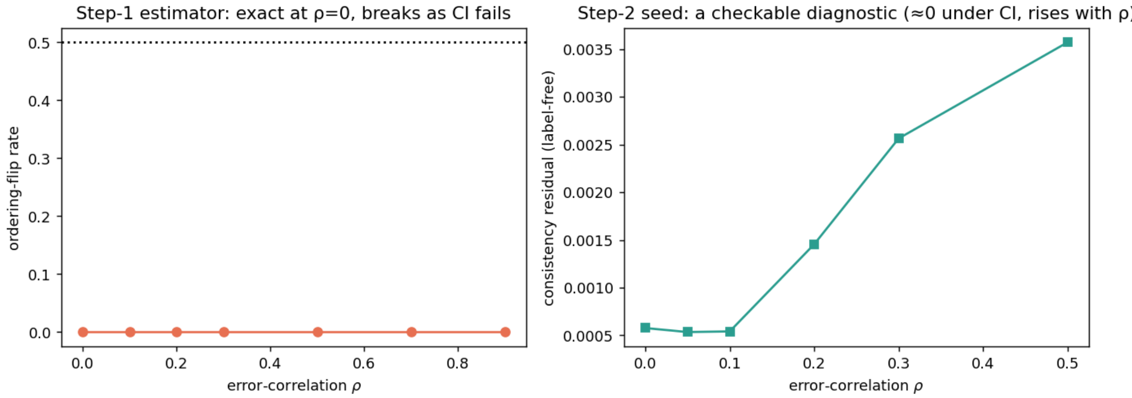

Figure 15: Multi-candidate identifiability (Theorem 16, Proposition 5). Left: the label-free ordering estimator is exact under conditional error-independence and degrades as it fails. Right: the overdetermination residual is a checkable diagnostic — ≈0 under independence, rising with error-correlation.

sign Δ under a condition that is, crucially, *testable* (Proposition 5) — the guarantee that agreement-based heuristics (ATC, AoL, AETTA) use empirically but never establish. It is not an assumption-free result: it substitutes conditional error-independence (often violated by co-trained models, which is exactly why the diagnostic matters) for the earlier crutches. The assumption-free dichotomy — the weakest checkable condition with a tight matching converse — remains open and is the stated frontier of this program (Conjecture 1).

C.7 The knowability frontier

Definition 6 (Knowability margin)**.** Fix a confidence level α and a label-free estimator $\widehat{\Delta}$ with a valid radius ε_α as in Theorem 4. The *knowability margin* at evidence z is

$$\kappa_\alpha(z) \;:=\; |\Delta(z)| \;-\; 2\,\varepsilon_\alpha(z).$$

$\kappa_\alpha > 0$ means the true benefit clears twice the certified uncertainty: the instance is *knowable at level* α.

Theorem 17 (Knowability frontier)**.** *Let* $\mathrm{FA} = \mathbb{P}[\textit{ADAPT} \wedge \Delta \le 0]$, $\mathrm{FF} = \mathbb{P}[\textit{FREEZE} \wedge \Delta \ge 0]$, *and coverage* $C = \mathbb{P}[\textit{ADAPT} \vee \textit{FREEZE}]$*.* (i) (Achievability.) *The certificate rule of Theorem 4 satisfies*

$$\mathrm{FA} \le \alpha, \qquad \mathrm{FF} \le \alpha, \qquad C \;\ge\; \mathbb{P}\big[\kappa_\alpha(Z) > 0\big] - \alpha.$$

(ii) (Converse.) *Call* z (δ, t)-ambiguous *if it belongs to a pair of admissible target worlds with evidence-law total variation at most* t *and opposite-sign benefits of magnitude at least* δ*. Any label-free rule whose wrong-commit probability is at most* α *in every admissible world commits on a* (δ, t)*-ambiguous pair with probability at most* $2\alpha + t$*. In particular, for observationally equivalent pairs* ($t = 0$)*, ambiguous evidence can be covered with probability at most* 2α*.*

(Proof deferred to Appendix A.)

Proposition 6 (Frontier sandwich)**.** *If the deployment distribution places mass* U *on* (δ, t)*-ambiguous evidence, then the optimal coverage at safety level* α *obeys*

$$\mathbb{P}[\kappa_\alpha(Z) > 0] - \alpha \;\le\; C^\star(\alpha) \;\le\; (1 - U) + (2\alpha + t)\,U.$$

Numerically validated (`knowability_frontier_validation.py`). *Synthetic* (planted ambiguous mass U=0.30, δ=0.4): across $\alpha \in [0.02, 0.30]$, $\mathrm{FA} \le \alpha$ and $\mathrm{FF} \le \alpha$ everywhere, coverage dominates the $\mathbb{P}[\kappa_\alpha > 0] - \alpha$ bound, and the commit rate *on the ambiguous subset* never exceeds the converse cap 2α — both directions of

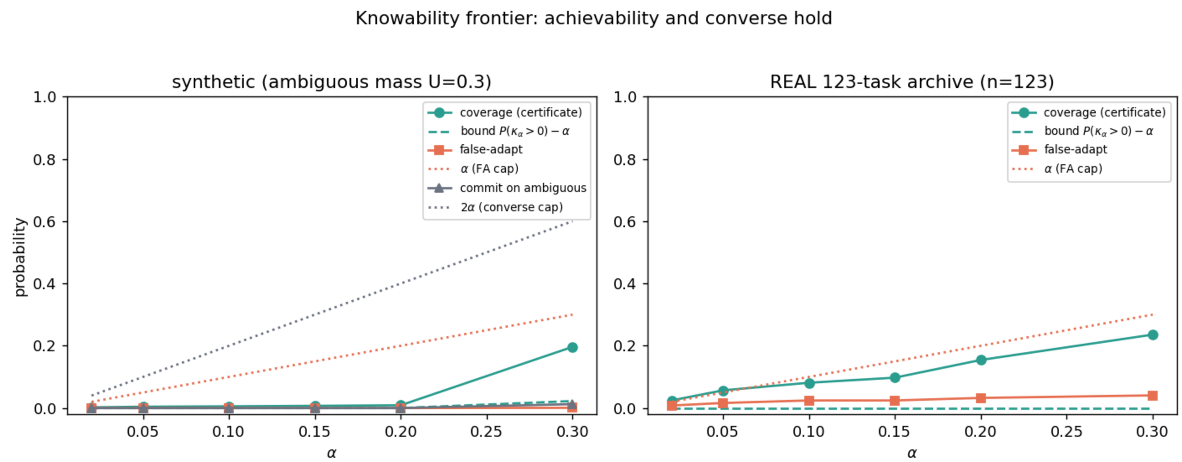

Figure 16: The knowability frontier (Theorem 17). Left: synthetic stream with planted ambiguous mass — false-adapt stays below α, coverage dominates the $\mathbb{P}[\kappa_\alpha > 0] - \alpha$ bound, and commits on ambiguous evidence respect the 2α converse cap. Right: the same certificate on the *real* 123-task archive. Produced by `scripts/knowability_frontier_validation.py`.

Theorem 17 hold (Fig. 16, left). *Real data* (the full 123-task score archive): the certificate keeps FA $\leq \alpha$ at every level (e.g. 0.024 at α=0.10; 0.016 at α=0.05) with coverage rising $0.02 \to 0.24$ as α goes $0.02 \to 0.30$ (Fig. 16, right). On this helpful-dominated suite $|\Delta|$ is small relative to $2\varepsilon_\alpha$, so the margin lower bound is loose (trivial) there — reported as is.

Prior decision styles as degenerate points on the frontier. The frontier also formalizes the positioning table: existing TTA decision styles are special — and weaker — points of the same object.

Proposition 7 (Always-adapt is the $\varepsilon \equiv 0$ certificate)**.** *Always-adapt is the rule of Theorem 4 with $\varepsilon \equiv 0$ and a positive prior estimate (it is κ-blind). Its regret equals the full false-adapt mass $\mathbb{E}[|\Delta|\mathbf{1}\{\Delta < 0\}]$ (Theorem 3), and on a $(\delta, 0)$-ambiguous pair its worst-case committal regret is δ — twice the Le Cam floor $\delta/2$ of Proposition 1 — while the certificate abstains (zero committal regret).*

(Proof deferred to Appendix A.)

Proposition 8 (Confidence filters lack false-adapt control)**.** *Let g commit* ADAPT *whenever a label-free confidence statistic exceeds a threshold (the decision style of entropy filtering). There exist confidently-wrong worlds — harmful and helpful instances with identical confidence laws — in which* FA(g) *equals the harmful base rate, while the certificate keeps* FA $\leq \alpha$ *on the same stream. The certificate's protection is purchased precisely by the calibrated benefit sample behind ε_α, not by any heuristic on Z.*

(Proof deferred to Appendix A.)

Proposition 9 (The decision is ordinal: it survives symmetric label noise)**.** *On the disagreement region D, let labels be corrupted by unknown symmetric noise $\eta < \frac{1}{2}$. Observed accuracies obey $a' - \frac{1}{2} = (1-2\eta)(a - \frac{1}{2})$, hence* $\operatorname{sign}(a'_a - a'_0) = \operatorname{sign}(a_a - a_0)$*: the adapt/freeze decision of Theorems 6–21 is invariant to η, while the absolute risks $R_T(f_0), R_T(f_a)$ are confounded by η and not identifiable. Label-free* accuracy estimation *(the AETTA/ATC target) therefore solves a strictly harder problem than the decision requires.*

(Proof deferred to Appendix A.)

Numerically validated. S1: on a mixed synthetic stream the always-adapt regret is 455.3 versus 0.0 committal regret for the certificate; on the witness pair, 1.0 versus the 0.5 floor. S2: in a confidently-wrong world the filter's false-adapt rate is 0.501 (the harmful share) versus $0.000 \leq \alpha$=0.1 for the certificate. S3: the decision sign is preserved at every $\eta \in \{0, \ldots, 0.4\}$ while the cardinal accuracy bias grows to 0.096 at η=0.4 (Fig. 17).

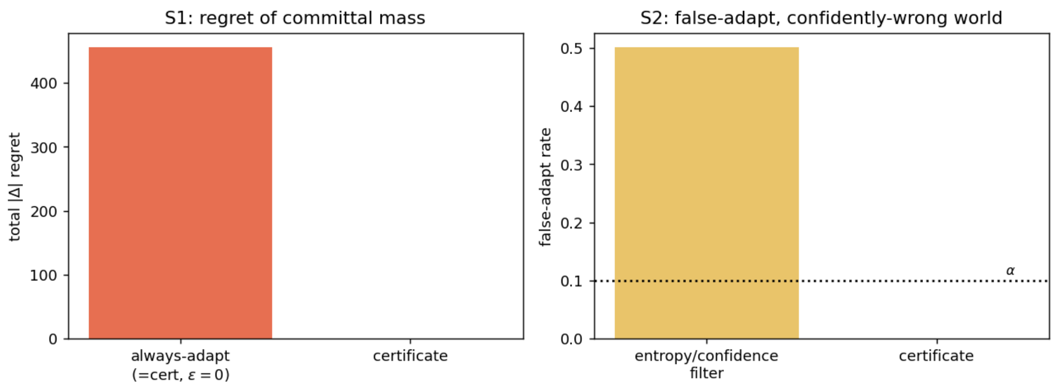

Figure 17: Prior decision styles as degenerate certificates (Propositions 7–8). Left: always-adapt pays the full false-adapt regret mass; the certificate's committal regret is zero. Right: in a confidently-wrong world an entropy/confidence filter adapts blindly while the certificate holds FA $\leq \alpha$.

Weight (honest). Theorem 17 assembles this paper's own ingredients — the certificate (Theorem 4), the Le Cam bound (Theorem 2), and the abstention logic — into a single quantitative frontier; the new elements are the margin primitive κ_α and the total-variation-slack converse. Propositions 7–9 formalize the positioning claims rather than introduce machinery. The synthetic stream validates both directions of the frontier including the 2α converse cap; the real-archive panel validates achievability, with the margin bound loose there because that suite is helpful-dominated.

C.8 How fast does knowability arrive: matching rates

Fix budgets α (validity) and β (abstention). Call a label-free rule (α, β)-*reliable at margin* κ if its wrong-commit probability is at most α in every admissible world, and it abstains with probability at most β whenever $|\Delta| \geq \kappa$. The question of this subsection: *how fast, in the number n of calibrated benefit samples, does the smallest reliable margin shrink?*

Theorem 18 (Knowability rate: adaptive achievability)**.** *Let $X_1, \dots, X_n \in [-R/2, R/2]$ be i.i.d. paired benefits with mean Δ and variance σ^2. The empirical-Bernstein certificate (Theorem 4 instantiated with the Maurer–Pontil radius, exactly as deployed in `switching_certificate.py`) is (α, β)-reliable at every margin*

$$\kappa \;\geq\; C\left(\sigma\sqrt{\frac{\log(2/(\alpha\wedge\beta))}{n}} \;+\; R\,\frac{\log(2/\alpha)}{n}\right),$$

for a universal constant C, without knowledge of σ*: the empirical variance self-normalizes the radius.*

(Proof deferred to Appendix A.)

Theorem 19 (Knowability rate: two-point lower bound)**.** *Let $\alpha + \beta < \frac{1}{4}$. Every label-free rule that is (α, β)-reliable at margin κ over all distributions on $[-R/2, R/2]$ with variance at most σ^2 must satisfy*

$$\kappa \;\geq\; c\,\max\left(\sigma\sqrt{\frac{\log\frac{1}{4(\alpha+\beta)}}{n}},\;\; R\,\frac{\log\frac{1}{2(\alpha+\beta)}}{n}\right)$$

for a universal constant $c > 0$ (one may take $c = \frac{1}{4}$ in the second branch and $c = \frac{1}{2\sqrt{2}}$ in the first).

(Proof deferred to Appendix A.)

Proposition 10 (The deployed certificate is rate-optimal and adaptive)**.** *The upper and lower rates match in both regimes up to universal constants: $\kappa_n^{\mathrm{EB}} \asymp \sigma\sqrt{\log(1/\bar{\alpha})/n} + R\log(1/\bar{\alpha})/n \asymp \kappa_n^{\mathrm{LB}}$. Consequently the knowability frontier of Theorem 17 is attained at the optimal rate by the certificate already in deployment, adaptively in σ.*

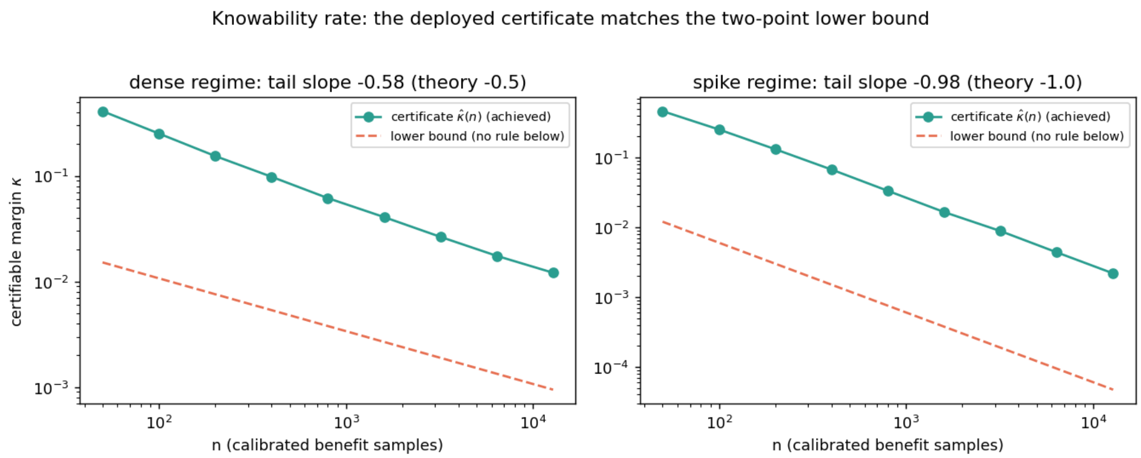

Figure 18: The knowability rate (Theorems 18–19). Achieved certifiable margin of the deployed empirical-Bernstein certificate versus the two-point lower bound, in the dense ($n^{-1/2}$) and spike (n^{-1}) regimes; slopes match and the offset is a bounded universal constant.

Numerically validated (`knowability_rates_validation.py`; Fig. 18). The vectorized radius coincides with the vendored implementation to 2×10^{-16}. The empirical smallest certifiable margin $\hat{\kappa}(n)$ has tail slope -0.59 in the dense regime (theory $-\frac{1}{2}$) and -0.98 in the spike regime (theory -1); its ratio to the lower bound is *constant in* n (within $\times 1.19$ and $\times 1.07$ over the tail), i.e. the rates match and the gap is a bounded constant. Wrong-commit at the achieved margin stays below α throughout. Below half the lower bound the *exact optimal* two-point error sum already exceeds the (α, β) budget at every n tested — no rule, certificate or otherwise, is reliable there.

Weight (honest). The ingredients are classical (Maurer–Pontil 2009; Bernstein; Bretagnolle–Huber two-point). The contribution is the (α, β)-reliability formulation of the adapt/freeze/abstain decision, the *two-regime matching characterization* of its sample complexity, and the fact that the certificate already deployed in this paper attains it *adaptively*. What remains open — and is not claimed — are rates for evidence-channel decisions (deciding from $Z = \phi(X_{1:n})$ over nonparametric shift families rather than from calibrated benefits), where only the location-family instance of Theorem 2 is currently characterized.

D Reliability-gated fusion as a worked, code-validated instantiation (T1–T9, GDR)

The general trichotomy (Theorems 1–6) is instantiated for reliability-gated multimodal fusion by a stack of ten results, each with code and a validator in the repository (vendored under `the supplementary theory code`). They are *supporting / instantiation* results—not the general trichotomy—and several are explicit operationalizations (e.g. CLT/Gaussian approximations in T3, partial real-validation in GDR); scope caveats are stated in each module. Their value here is concrete: they show the abstract boundary has a fully worked, machine-checked special case. The decisive mappings are **T5 = Theorem 4** (the finite-sample certificate), **T9 = the proven knowably-harmful/clean regime**, and **GDR = minimax justification of the abstain action** (Theorem 1).

ID	One-line statement	K-Bound role / regime
T1	Reliability-*blind* fusion is dominated under sparse/coherent corruption (positive lower bound).	Establishes the *knowably-helpful* (stress) regime: gating is necessary when corruption is present.
T2	Global KS drift is inflated by benign mixture re-weighting even with no real drift.	Observability hazard: evidence Z can *false-fire* $\Rightarrow$ a route into false-adapt / unknowable.
T3	Closed-form mean-gate miss probability (CLT approx.).	Quantifies gate error; calibrates the certificate.
T4	Finite-sample risk-dominance sample complexity for the switch.	How much validation certifies the helpful regime.
T5	Empirical-Bernstein finite-sample switching certificate.	**Instantiates Theorem 4** (false-adapt-controlled rule).
T6	Windowed-KS gate as a CUSUM detector; ARL/AED trade-off.	Detectability of the shift trigger: when Z is informative.
T7	Massart finite-class Rademacher bound for the meta-router.	Generalization of the benefit estimator $\widehat{\Delta}$ in Theorem 4.
T8	Validation-only certified selection among {gate, TTA, stack}.	Multi-candidate KGA decision (beyond binary adapt/freeze).
T9	No fusion rule beats the confidence-weighted mean on clean near-separable transfer.	**Proves the knowably-harmful/neutral (clean) regime** $\Rightarrow$ freeze is optimal.
GDR	Coherence-certified switch is minimax over coherent-shift vs heterogeneous-mixture adversaries.	**Justifies the abstain/gate action** as minimax-safe (Theorem 1).

Together, T1/T3/T4 lower-bound the *helpful* regime, T9 proves the *harmful/neutral* regime, T2/T6/T7 govern the *observability* of the evidence Z (and hence the boundary of the *unknowable* regime), T5 supplies the operational certificate, and GDR certifies abstention. This is the fusion-case shadow of the general {adapt, freeze, abstain} theory.

E Anytime-valid certificate (testing by betting)

Theorem 4 fixes the sample size n in advance. For streaming TTA we give a sequential complement that is valid at *any* stopping time (no multiplicity correction for repeated looks), via an e-process.

Theorem 20 (Anytime-valid adapt/freeze/abstain certificate)**.** *Let paired benefits $X_i = \ell(f_0(x_i), y_i) - \ell(f_a(x_i), y_i) \in [a, b]$ ($a < 0 < b$) be adapted to $(\mathcal{F}_t)$ with $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = \Delta$. With predictable bets $\lambda_t \in [0, c/(-a)]$, $\nu_t \in [0, c/b]$, $c \in (0, 1)$, set $E_t^+ = \prod_{i \le t}(1 + \lambda_i X_i)$ and $E_t^- = \prod_{i \le t}(1 - \nu_i X_i)$. Decide* ADAPT *the first time $E_t^+ \ge 1/\alpha$,* FREEZE *the first time $E_t^- \ge 1/\alpha$, else* ABSTAIN*. Then simultaneously over all t, $\mathbb{P}[\exists t : E_t^+ \ge 1/\alpha \wedge \Delta \le 0] \le \alpha$ and $\mathbb{P}[\exists t : E_t^- \ge 1/\alpha \wedge \Delta \ge 0] \le \alpha$.*

(Proof deferred to Appendix A.) A predictable truncated growth-optimal bet $\lambda_t = \mathrm{clip}(\widehat{\mu}_{t-1}/\widehat{\sigma}^2_{t-1}, 0, c/(-a))$ governs power; validity needs only predictability and boundedness. *Numerical sanity check* (`val_thm3_evalue.py`): anytime false-adapt $0.06 \le \alpha$=0.1 (and $0.032 \le 0.05$), power $\to 1$ with mean stopping time shrinking as $|\Delta|$ grows. It *complements* (does not replace) the batch certificate: tighter at a fixed n, vs valid at any stopping time.

F Extended theory: variants, boundaries, capacity, and drift

F.1 Disagreement-region variants (multiclass and regression)

Theorem 21 (Multiclass 0/1 loss)**.** *For $Y \in \{1, \ldots, K\}$ with 0/1 loss, with $p_a = P_T(f_a{=}Y \mid D)$, $p_0 = P_T(f_0{=}Y \mid D)$, off D $\delta = 0$ and on D $\mathbb{E}[\delta \mid D] = p_a - p_0$, so $\Delta = P_T(D)(p_a - p_0)$ and* $\operatorname{sign}\Delta = \operatorname{sign}(p_a - p_0)$.

(Proof deferred to Appendix A.)

Theorem 22 (Squared-error regression)**.** *For $Y \in \mathbb{R}$ with squared loss, $\delta = (f_0 - f_a)(f_0 + f_a - 2Y)$, so on D $\mathbb{E}[\delta \mid D] = \mathbb{E}[(f_0 - f_a)(f_0 + f_a) \mid D] - 2\,\mathbb{E}[(f_0 - f_a)Y \mid D]$. The first term is observable from unlabeled target; thus* $\operatorname{sign}\Delta$ *is label-free identifiable iff the sign contribution of $\mathbb{E}[(f_0 - f_a)Y \mid D]$ is certifiable—which holds under covariate shift (importance-weighted source estimate of $\mathbb{E}[Y \mid X]$) and fails under concept shift (re-entering Theorem 1).*

(Proof deferred to Appendix A.) Numerical sanity check (`val_thm5_multiclass.py`): the multiclass identity holds to 10^{-16} with 100% sign agreement over 4000 trials (with both predictors wrong on D in every trial, confirming the beyond-binary regime); the regression certificate recovers the correct sign under covariate shift and flips under concept shift.

F.2 A partial resolution of Conjecture 1: the bounded-drift boundary

G All 65 CIFAR-10-C cells (per-corruption × severity)

Table 10 lists every cell behind the summary in Table 6 (KGA wraps Tent; EATA/SAR are the standalone candidates). The per-method means reproduce Table 6 exactly. Raw CSV: `experiments/kbound/results/cifar10c_65cells.cs`

Table 10: All 65 CIFAR-10-C cells (13 official corruptions × severities 1–5), ResNet-18, N=800 balanced held-out eval per cell. Online TTA accuracy; KGA wraps Tent. Mean row matches Table 6.

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
brightness	1	0.863	0.858	0.856	0.859	0.858	0.865
brightness	2	0.844	0.864	0.864	0.865	0.864	0.868
brightness	3	0.804	0.837	0.837	0.839	0.837	0.839
brightness	4	0.731	0.818	0.818	0.815	0.819	0.816
brightness	5	0.690	0.791	0.794	0.795	0.791	0.799
contrast	1	0.719	0.831	0.831	0.835	0.831	0.831
contrast	2	0.559	0.823	0.821	0.822	0.820	0.820
contrast	3	0.325	0.790	0.793	0.793	0.790	0.791
contrast	4	0.223	0.728	0.729	0.730	0.726	0.726
contrast	5	0.160	0.611	0.607	0.609	0.614	0.601
defocus_blur	1	0.264	0.683	0.680	0.681	0.682	0.676
defocus_blur	2	0.207	0.543	0.544	0.544	0.540	0.540
defocus_blur	3	0.204	0.389	0.386	0.398	0.389	0.385
defocus_blur	4	0.176	0.300	0.300	0.303	0.300	0.305
defocus_blur	5	0.183	0.265	0.264	0.261	0.261	0.275
elastic_transform	1	0.244	0.528	0.528	0.530	0.528	0.525
elastic_transform	2	0.179	0.436	0.440	0.446	0.436	0.452
elastic_transform	3	0.168	0.401	0.401	0.405	0.401	0.400
elastic_transform	4	0.144	0.407	0.407	0.413	0.407	0.410
elastic_transform	5	0.151	0.347	0.346	0.353	0.351	0.351
frost	1	0.725	0.804	0.805	0.806	0.804	0.796
frost	2	0.588	0.760	0.762	0.761	0.761	0.762
frost	3	0.546	0.685	0.685	0.686	0.684	0.697
frost	4	0.492	0.705	0.705	0.704	0.705	0.706
frost	5	0.436	0.640	0.642	0.644	0.640	0.659
gaussian_noise	1	0.487	0.741	0.740	0.741	0.739	0.739
gaussian_noise	2	0.275	0.699	0.699	0.700	0.700	0.701
gaussian_noise	3	0.144	0.594	0.595	0.595	0.593	0.592
gaussian_noise	4	0.100	0.435	0.435	0.432	0.436	0.441
gaussian_noise	5	0.089	0.305	0.304	0.301	0.305	0.301
impulse_noise	1	0.615	0.769	0.770	0.769	0.769	0.775
impulse_noise	2	0.370	0.682	0.682	0.682	0.682	0.688

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
impulse_noise	3	0.197	0.605	0.605	0.604	0.608	0.623
impulse_noise	4	0.110	0.475	0.479	0.485	0.476	0.484
impulse_noise	5	0.082	0.335	0.335	0.335	0.335	0.331
jpeg_compression	1	0.676	0.730	0.730	0.731	0.730	0.735
jpeg_compression	2	0.639	0.660	0.659	0.661	0.662	0.663
jpeg_compression	3	0.608	0.655	0.654	0.656	0.654	0.658
jpeg_compression	4	0.509	0.573	0.570	0.570	0.573	0.567
jpeg_compression	5	0.460	0.501	0.501	0.501	0.501	0.500
motion_blur	1	0.495	0.782	0.784	0.785	0.784	0.783
motion_blur	2	0.365	0.666	0.665	0.664	0.666	0.675
motion_blur	3	0.279	0.575	0.574	0.575	0.574	0.590
motion_blur	4	0.255	0.490	0.491	0.489	0.491	0.502
motion_blur	5	0.225	0.456	0.450	0.446	0.456	0.476
pixelate	1	0.429	0.719	0.720	0.721	0.716	0.720
pixelate	2	0.373	0.651	0.651	0.654	0.650	0.654
pixelate	3	0.306	0.570	0.574	0.576	0.573	0.571
pixelate	4	0.267	0.487	0.485	0.487	0.489	0.486
pixelate	5	0.289	0.405	0.404	0.406	0.405	0.410
shot_noise	1	0.492	0.779	0.776	0.779	0.777	0.784
shot_noise	2	0.254	0.654	0.655	0.659	0.654	0.669
shot_noise	3	0.155	0.536	0.537	0.549	0.534	0.545
shot_noise	4	0.125	0.400	0.398	0.398	0.398	0.404
shot_noise	5	0.120	0.303	0.305	0.306	0.300	0.300
snow	1	0.770	0.782	0.781	0.779	0.782	0.781
snow	2	0.578	0.686	0.685	0.685	0.685	0.690
snow	3	0.661	0.716	0.716	0.711	0.718	0.710
snow	4	0.574	0.683	0.681	0.679	0.683	0.680
snow	5	0.459	0.597	0.596	0.597	0.597	0.598
zoom_blur	1	0.760	0.877	0.879	0.876	0.876	0.879
zoom_blur	2	0.708	0.847	0.847	0.849	0.847	0.853
zoom_blur	3	0.674	0.830	0.829	0.828	0.829	0.826
zoom_blur	4	0.635	0.809	0.810	0.810	0.807	0.814
zoom_blur	5	0.557	0.779	0.776	0.778	0.780	0.785
mean		**0.412**	**0.626**	**0.626**	**0.627**	**0.626**	**0.629**

H Proof and experiment status

Theorem 1 is proved with a constructed witness and strengthened to a quantitative Le Cam minimax bound (Theorem 2); Theorem 3 is a plug-in regret decomposition (exact identity) with a minimax floor; Theorem 4 is proved *conditional* on a valid label-free interval estimator and complemented by an anytime-valid e-process (Theorem 20); Theorem 5 proves the covariate case and Theorems 6–22 the binary, multiclass, and regression sign-of-difference results. The multiclass/regression label-free *bracketing* is closed under a bounded calibration-transfer assumption (Theorem 9); the *unconditional* bracketing (Conjecture 1) is the one residual open piece. Every theorem ships a numerical sanity-check script (`experiments/kbound/theory_validation/`); all were re-run from a clean environment and pass: the Le Cam floor tracks the closed form, the regret identity holds to $\sim 10^{-17}$, the multiclass identity to 10^{-16} with 100% sign agreement over 4000 trials, calibration-transfer sign-recovery is 100% when the confidence gap exceeds 2η, the router generalization gap decays as $\sim n^{-0.66}$, and the drift-corrected anytime false-adapt rate stays $\leq \alpha$. Where multi-seed runs are available, experiments report means $\pm$ std with paired t-tests; single-seed/scoping tracks are labeled explicitly in captions and discussed as regime-classification evidence. We also report evidence/α/batch ablations, a regression covariate-shift track, and a clean non-identifiability witness.